POLYTECHNIQUE MONTRÉAL
affiliée à l'Université de Montréal

A lightweight canonicalization engine in C for disciplined convex programming

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Mémoire présenté en vue de l'obtention du diplôme de *Maîtrise ès sciences appliquées*
Electrical Engineering

July 2026

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Ce mémoire intitulé :

A lightweight canonicalization engine in C
for disciplined convex programming

présenté par **William ZHANG**
en vue de l'obtention du diplôme de *Maîtrise ès sciences appliquées*
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DEDICATION

To all the past, present and future CVXPY contributors

ACKNOWLEDGEMENTS

Texte / Text.

RÉSUMÉ

Le résumé est un bref exposé du sujet traité, des objectifs visés, des hypothèses émises, des méthodes expérimentales utilisées et de l'analyse des résultats obtenus. On y présente également les principales conclusions de la recherche ainsi que ses applications éventuelles. En général, un résumé ne dépasse pas trois pages.

Le résumé doit donner une idée exacte du contenu du mémoire ou de la thèse. Ce ne peut pas être une simple énumération des parties du manuscrit. Le but est de présenter de façon précise et concise la nature, l'envergure de la recherche, les sujets traités, les questions de recherche ou les hypothèses soulevées, les méthodes utilisées, les principaux résultats ainsi que les conclusions retenues. Un résumé ne doit jamais comporter de références ou de figures.

ABSTRACT

Written in English, the abstract is a brief summary similar to the previous section (Résumé). However, this section is not a word for word translation of the abstract in French.

The abstract is a brief statement of the subject matter, objectives, research questions or hypotheses, experimental methods and analysis of results. It also presents the main research conclusions and their possible applications. In general, an abstract should not exceed three pages.

The abstract should provide an exact idea of the thesis or dissertation's contents and it cannot be a simple enumeration of the manuscript's parts. The goal is to precisely and concisely present the nature and scope of the research. An abstract should never include references or figures. If the thesis or the dissertation is in English, the résumé (French-language abstract) should come first followed by the abstract.

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LIST OF SYMBOLS AND ACRONYMS

AD	Algorithmic Differentiation
AML	Algebraic Modeling Language
API	Application Programming Interface
ASA	Affine-Solver-Affine
ASL	AMPL Solver Library
BLAS	Basic Linear Algebra Subprograms
CSC	Compressed Sparse Column
CSR	Compressed Sparse Row
DAG	Directed Acyclic Graph
DCP	Disciplined Convex Programming
DPP	Disciplined Parameterized Programming
LP	Linear Program
MOI	MathOptInterface
MPS	Mathematical Programming System
NLP	Nonlinear Program
QP	Quadratic Program
SOCP	Second-Order Cone Program
SDP	Semidefinite Program

CHAPTER 1 INTRODUCTION

Since the early days of linear programming, practitioners of optimization have needed to translate mathematical formulations into data that solvers can process. At first, this meant encoding each coefficient of the problem matrices by hand, a process that can both be tedious and error-prone [1]. *Matrix generators* were introduced in the late 1960s to automate this task: they compute the coefficients through explicit loops and write them out in solver-readable formats like MPS [2, 3].

Matrix generators were able to remove most of the manual labor, but a gap still remained. Modelers naturally think in terms of sets, indices, and algebraic expressions, while procedural code constructs the coefficient matrices entry by entry. This mismatch between how a model is expressed and how it is implemented made them difficult to verify, modify, and communicate with others [4]. *Algebraic modeling languages* such as `AMPL` [5], `AIMMS` [6] and `GAMS` [7] closed this gap by accepting problem descriptions that closely follow the math. The survey by Orchard-Hays [8] gives a more complete overview of the early history of modeling languages.

More recently, modeling languages embedded in general-purpose programming languages have gained popularity. These include `Pyomo` [9,10] in Python, `JuMP` [11] in Julia, `CasADi` [12] in C++, and `YALMIP` [13] in MATLAB. Other domain-specific modeling languages specialized for convex optimization [14] are built on *disciplined convex programming* (DCP), the composition rules under which the convexity of a problem can be verified automatically [15, 16]. These DCP-based languages include `CVX` [17] in MATLAB, `CVXR` [18] in R, `CVXPY` [19] in Python, and `Convex.jl` [20] in Julia.

In the meantime, modern general purpose solvers [21–23] have mostly transitioned towards conic programming which generalizes linear programming and can represent nearly every convex program used in practice [24]. The translation task remains a responsibility of the modeling languages which now target conic problem data instead of LP coefficient matrices. In DCP-based languages, this translation process is called *canonicalization*. While the time taken to solve an optimization problem is often dominated by the solver, the time taken to *canonicalize* (or *compile*) it can also be significant. The main goal of this thesis is to reduce the compilation time of `CVXPY`.

1.1 Problem statement

The standard form of a convex optimization problem is

$$\begin{array}{ll} \text{minimize} & f_0(x;\theta) \\ \text{subject to} & f_i(x;\theta) \leq 0, \quad i = 1,\ldots,m_1 \\ & g_j(x;\theta) = 0, \quad j = 1,\ldots,m_2 \end{array} \tag{1.1}$$

where $x \in \mathbf{R}^n$ is the optimization variable, the objective f_0 and inequality constraints $f_1,\ldots,f_{m_1}$ are convex and the equality constraints $g_1,\ldots,g_{m_2}$ are affine [14]. The parameter $\theta \in \mathbf{R}^p$ specifies data that can change between solves but is fixed and given when solving an instance of the problem. We call the parametrized problem (1.1) a problem *family*; for a fixed value of θ, we call it a problem *instance*.

Quadratic conic form. Canonicalization converts the high-level description (1.1) to a *quadratic conic form* used by many general-purpose convex solvers

$$\begin{array}{ll} \text{minimize} & \frac{1}{2}x^T P(\theta)x + c(\theta)^T x \\ \text{subject to} & b(\theta) - A(\theta)x \in K \end{array} \tag{1.2}$$

where the *problem data* are the symmetric positive semidefinite matrix $P(\theta) \in \mathbf{S}^n_+$, the matrix $A(\theta) \in \mathbf{R}^{m\times n}$, the vectors $b(\theta) \in \mathbf{R}^m$ and $c(\theta) \in \mathbf{R}^n$ and the non-empty closed convex cone $K \subseteq \mathbf{R}^m$. Problem (1.2) generalizes both quadratic programs (QPs), when $K = \mathbf{R}^m_+$ is the nonnegative orthant, and standard conic programs, when $P(\theta) = 0$.

Canonicalization in `CVXPY`. Most of the cost of canonicalization lies in the final step, where symbolic expression trees are converted to numerical problem data. This step is often called *matrix stuffing* or the *backend* of canonicalization. During the first solve of a problem family, `CVXPY` forms and caches a *parameter tensor* that maps the parameter vector to the problem data. On subsequent solves, new problem data can be obtained with a single matrix-vector product [25].

For large-scale problems, forming the parameter tensor can be prohibitively expensive, so `CVXPY` lets users opt out of this option entirely. Users must thus choose between paying a large upfront cost on the first solve and repeating the full canonicalization, with nothing reused, at every solve. We review `CVXPY` and the frontend of canonicalization in §1.3, as well as the parameter tensor and matrix stuffing in §2.1.

Matrix stuffing as a derivative. The constraint matrix A is the Jacobian of the stacked constraint expressions and P is the Hessian of the objective, so matrix stuffing amounts to computing a sparse Jacobian and Hessian. Classical methods build these matrices column by column with multiple passes of algorithmic differentiation (AD), using graph coloring to group columns that can be recovered in a single pass [26, 27]. These methods treat the expression as a black box: the sparsity pattern must first be detected and the columns grouped before a single entry of the matrix is obtained. We review the basics of computing derivatives in §1.4 and present sparse AD and graph coloring methods in §2.2.

Other approaches. A modeling language, by contrast, knows which variables enter each constraint. `AMPL` [5] is split in two components: its commercial *translator* writes the structure of the model to a `.nl` file [28], and the open source AMPL Solver Library (ASL) evaluates constraints and their derivatives from this file inside the solver process [29]. `Pyomo` [9] also interfaces to solvers through `.nl` files, among other formats, though it generally writes them more slowly than the `AMPL` translator. On the other hand, `JuMP` [11] has no single standard form to target; it decouples modeler and solver through MathOptInterface (MOI), a typed in-memory interface [30]. The model is built constraint by constraint, and each constraint stores its coefficients directly, so extraction reduces to copying them into the format of the chosen solver. None of these approaches operates on expression trees: the AD tools record the expression as a flat tape of instructions, `AMPL` exchanges the model through a file, and `JuMP` stores lists of coefficients. In `CVXPY` the expression trees themselves are the input of the backend. We review both these modeling languages in §2.3.

Our approach. We make use of the fact that the sparsity patterns of A and P are fixed across a problem family, since the parameter θ changes the values of the problem data but not their structure. We can therefore split matrix stuffing in a symbolic and numeric phase. The symbolic phase runs once on the first solve, where it computes the sparsity patterns, allocates memory for all intermediate matrices, and caches other computations that remain valid across the problem family. The numeric phase then fills in new values at every solve. This separation between structure and values is inspired by the interfaces of nonlinear programming solvers such as `IPOPT` [31].

The symbolic phase is much cheaper to form and cache than the parameter tensor, and the numeric phase is often similar in cost to applying the matrix-vector product. The problem data is also assembled entirely in memory rather than through a separate model file. Finally, we exploit structure within the sparsity patterns themselves, which often contain dense blocks of data. When possible, we store these blocks as dense matrices and compute with dense

linear algebra kernels, rather than always relying on generic sparse formats and routines (we review dense and sparse matrix computations in §1.5).

1.2 Contributions

We make the following contributions:

1. We build `SparseDiffEngine`, a C library with Python bindings that computes sparse Jacobians and Hessians without graph coloring.
2. We implement a canonicalization backend for `CVXPY` built on `SparseDiffEngine`; this backend avoids the large upfront cost of the parameter tensor while still caching computations for subsequent solves.
3. We benchmark this backend against `CVXPY`'s existing canonicalization backends.

All code contributions are open source and available at

`https://github.com/SparseDifferentiation/SparseDiffengine`
`https://github.com/cvxpy/cvxpy`
`https://github.com/SparseDifferentiation/Jacobian-accumulation`

Thesis outline. The rest of this chapter (§1.3 to §1.5) presents some important background on canonicalization, computing derivatives, and matrix computations. §2 reviews the three existing methods: `CVXPY`'s matrix stuffing (§2.1), sparse derivatives by detection and coloring (§2.2), and problem data from declared structure in `AMPL` and `JuMP` (§2.3). In §3 we present an overview of the main data structures and algorithms in `SparseDiffEngine` as well as its interface in `CVXPY`. Finally, §4 benchmarks this engine against `CVXPY`'s three canonicalization backends, and concludes with a summary of our contributions and future improvements to the canonicalization backend in §5.

1.3 Canonicalization

The problem family (1.1) is written in *functional* form: an objective and constraint functions of the variable and the parameters. In a modeling language, the optimization variables $x_1, \ldots, x_k$ are collected into a single vector $x \in \mathbf{R}^n$, and likewise the parameters $\theta_1, \ldots, \theta_l$ into $\theta \in \mathbf{R}^p$ [14]. Users are not required to write constraints with a zero right-hand side as in (1.1): modeling languages accept inequalities with a convex left-hand side and a concave

right-hand side, and equalities with both sides affine. Internally, each function is represented as an *expression tree*.

Atoms. Modeling languages for convex optimization are built on top of a library of functions which can be used to describe problems. These functions are known as *atoms* in `CVXPY` since they are the fundamental building blocks of the modeling language [15]. Alongside its numerical definition, each atom declares useful attributes such as its *curvature*, *monotonicity* with respect to each argument, and the *sign* of its output. A selection of common atoms is shown in Table 1.1.

Table 1.1 A selection of `CVXPY` atoms with their DCP attributes: curvature, monotonicity in each argument, and sign of the output.

Atom	Definition	Curvature	Monotonicity	Sign
`sum`	$\sum_i x_i$	affine	non-decreasing	any
`abs`	$\lvert x\rvert$	convex	unknown	nonnegative
`square`	x^2	convex	unknown	nonnegative
`sqrt`	$\sqrt{x}$	concave	non-decreasing	nonnegative
`exp`	e^x	convex	non-decreasing	positive
`log`	$\log x$	concave	non-decreasing	any
`norm`	$\lVert x\rVert_2$	convex	unknown	nonnegative
`max`	$\max_i x_i$	convex	non-decreasing	any
`sum_largest`	sum of k largest entries	convex	non-decreasing	any

Expression trees. An expression is defined recursively as an atom evaluated at *subexpressions*, each of which is itself another expression. Such an expression is naturally stored as an expression tree, with atoms at its internal nodes and variables, parameters, and constants at its leaves. In Python, this tree can be formed automatically from user code through operator overloading. Most operations on an expression, from evaluation to verifying convexity to differentiation, traverse the tree and combine the intermediate results at each node.

Disciplined convex programming. When f_0 and all f_i are convex and all g_j are affine, (1.1) defines a convex optimization problem. To certify convexity, DCP-based modeling languages use the DCP composition rules to classify the curvature of each function [15]. Once all functions are classified, a problem is *DCP-compliant* if the objective and each constraint have curvatures compatible with a convex problem: a convex objective, convex $\leq$ concave inequalities, and affine equalities. Otherwise, if a curvature is incompatible (*e.g.* f_0 is concave) or cannot be determined, the modeling language raises an error and terminates. Because these

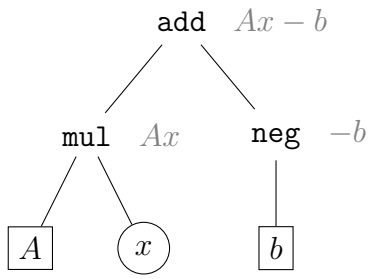


Figure 1.1 Example of an affine expression tree for $Ax - b$. Variables are drawn as circles and constants (or parameters) as boxes.

rules are only sufficient for convexity, a problem that is genuinely convex but not expressed in a recognized form may still be rejected.

Composition rules. The composition $\phi(f_1(x), f_2(x), \ldots, f_k(x))$ of an atom ϕ with subexpressions f_i is convex when ϕ is convex and, for each argument $i \in \{1, \ldots, k\}$, one of the following holds:

- ϕ is non-decreasing in argument i and f_i is convex, or
- ϕ is non-increasing in argument i and f_i is concave, or
- f_i is affine.

The same results hold for concave functions by swapping convex and concave. Affine expressions require ϕ and all of its arguments to be affine [15].

1.3.1 Conic reformulation

Once a problem is verified to be DCP-compliant, canonicalization rewrites its functional form (1.1) into an equivalent *functional conic form*

$$\begin{array}{ll} \text{minimize} & f_0(x;\theta) \\ \text{subject to} & h_i(x;\theta) \in \mathcal{K}_i, \quad i = 1, \ldots, m, \end{array} \tag{1.3}$$

where f_0 is affine or quadratic, each h_i is affine, and each $\mathcal{K}_i$ is one of the standard convex cones: the zero, nonnegative, second-order, power, exponential, and positive semidefinite cones.

Conic reformulation. Every nonlinear atom is rewritten into conic constraints in three steps. The first step rewrites the problem into *Smith form*, where each nonlinear atom is replaced by a new variable coupled with an equality constraint; every atom then acts only on affine arguments [32]. For a convex atom ϕ, the Smith form gives an auxiliary variable t with the equality constraint $\phi(x) = t$. The second step relaxes this equality to the inequality $\phi(x) \leq t$. The DCP rules allow a convex atom only where its value is minimized or upper-bounded, so the solver always prefers t as small as possible; the relaxation is therefore *lossless*.

The pair (x, t) then lies in the *epigraph* $\{(x, t) : \phi(x) \leq t\}$ of ϕ, so we often refer to t as an *epigraph variable*. A concave atom relaxes to $\phi(x) \geq t$ instead, giving a *hypograph variable*. The third and final step expresses each epigraph (hypograph) inequality as the membership of an affine function of (x, t) in a convex cone [33].

Example of reformulations. A first example is the ℓ_2 norm atom $\|x\|_2$, whose epigraph $\|x\|_2 \leq t$ is exactly the second-order cone constraint $(t, x) \in \mathcal{K}_{\text{soc}}$. A second example is the nondifferentiable absolute value atom $|x|$, whose epigraph $|x| \leq t$ can be written smoothly as the pair of affine inequalities $-t \leq x \leq t$. A final example is the sum of the k largest entries of x, which expands into a small linear program (LP): the epigraph sum_largest$(x, k) \leq t$ holds if and only if there exist a scalar q and a vector u such that

$$q\mathbf{1} + u \geq x, \qquad u \geq 0, \qquad kq + \mathbf{1}^\top u \leq t. \tag{1.4}$$

Here q acts as a threshold and each u_i measures how far the entry x_i rises above it, so that the sum of the k largest entries is bounded by $kq + \mathbf{1}^\top u$ [14].

1.3.2 `CVXPY`

`CVXPY` is a DCP-based modeling language embedded in Python [19], which began in 2013 as an implementation of the modeling principles of `CVX` [17]. Three fundamental design decisions differed from `CVX` at the beginning. The first is modularity: a few core data structures are shared by separate components for verification, canonicalization, matrix stuffing, and solver interfaces. The second is the expression tree as the central data structure, which is formed automatically from user code through Python's operator overloading. The third is support for parameters: since canonicalization never looks at the values of constants, much of the compilation work can be reused when a problem is solved repeatedly. These ideas matured with version 1.0, which redesigned canonicalization as a general reduction-based rewriting system [34]. The new design enabled the addition of other disciplined grammars [35–37], and the early parameter support grew into DPP, the restricted grammar under which

canonicalized programs can be updated efficiently [25].

The open source project has since grown through contributions from many other enthusiastic developers; a recent paper surveys the new canonicalization and modeling features introduced between versions 1.1 and 1.9 [38]. The exposition of canonicalization in this chapter follows this survey, with more emphasis on the non-DPP path.

Reductions. The rewriting from the functional form (1.1) to the functional conic form (1.3) is carried out as a sequence of *reductions*. Each reduction provides an `apply` step that converts a problem into an equivalent one, and a `retrieve` step that recovers a solution of the original problem from a solution of the rewritten one [34].

A first example is `FlipObjective`, which turns a maximization problem into a minimization one by flipping the sign of the objective; its `retrieve` step simply flips the sign back.

A second example, `CvxAttr2Constr`, handles variable attributes, which are properties declared on a variable that restrict its domain. These include nonnegativity (`nonneg`), positive semidefiniteness (`PSD`), and explicit variable bounds, amongst others. When the target solver supports an attribute natively, `CVXPY` leaves it for the solver to enforce; otherwise the reduction creates a new variable that carries no attributes and adds explicit constraints that enforce the attribute. Its `retrieve` step then maps the reformulated solution back onto the original attributed variable.

The `Dcp2Cone` reduction. The main reduction, `Dcp2Cone`, applies the epigraph reformulation of §1.3.1 atom by atom: the problem is first rewritten into Smith form, each resulting equality is relaxed to an epigraph (or hypograph) inequality, and each inequality is finally expressed as a cone membership [32, 33]. In practice the three steps are performed at once: each nonlinear atom is replaced directly by its epigraph variable and cone constraints, and the intermediate Smith form and its relaxation never appear explicitly.

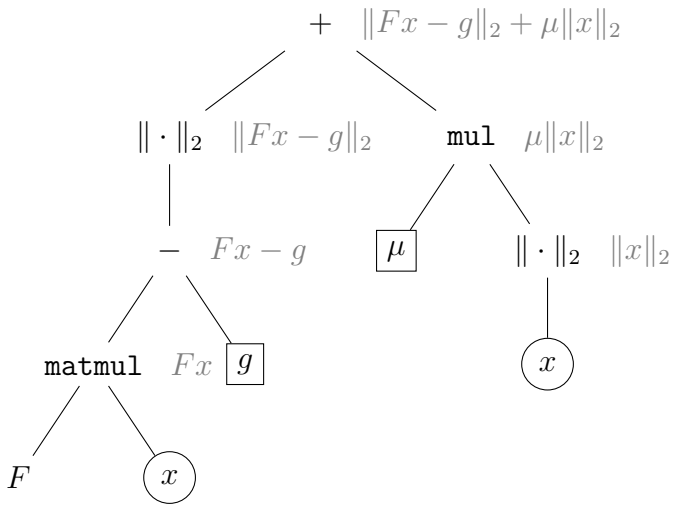
Example of canonicalization. To show a concrete example of the frontend of canonicalization, we consider a simple fitting with regularization problem

$$\begin{array}{ll} \text{minimize} & \|Fx - g\|_2 + \mu\|x\|_2 \\ \text{subject to} & x \succeq 0 \end{array} \tag{1.5}$$

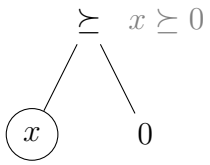
with variable $x \in \mathbf{R}^n$, fixed problem data $F \in \mathbf{R}^{m\times n}$, and parameters $g \in \mathbf{R}^m$ and $\mu \geq 0$; the constraint $x \succeq 0$ requires every entry of x to be nonnegative. The problem formulation in `CVXPY` code is

```
import cvxpy as cp
import numpy as np
F = np.random.randn(m, n)
x = cp.Variable(n)
g = cp.Parameter(m)
mu = cp.Parameter(nonneg=True)
objective = cp.Minimize(cp.norm(F @ x - g) + mu * cp.norm(x))
prob = cp.Problem(objective, [x >= 0])
```

where we also use the numerical computation library `NumPy` [39] for the data arrays. The expression tree representation of the problem is shown in Figure 1.2.



(a) Expression tree of the objective



(b) Expression tree of the constraints

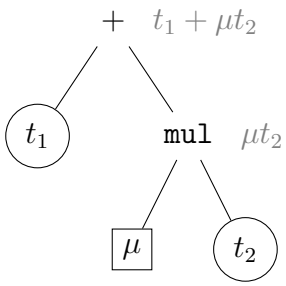
Figure 1.2 *Before canonicalization.* `CVXPY` represents the DCP problem (1.5) as these expression trees. Variables are labeled with a circle, and parameters with a square.

Conic form expression tree. After applying the reductions, the problem has been *canonicalized* to

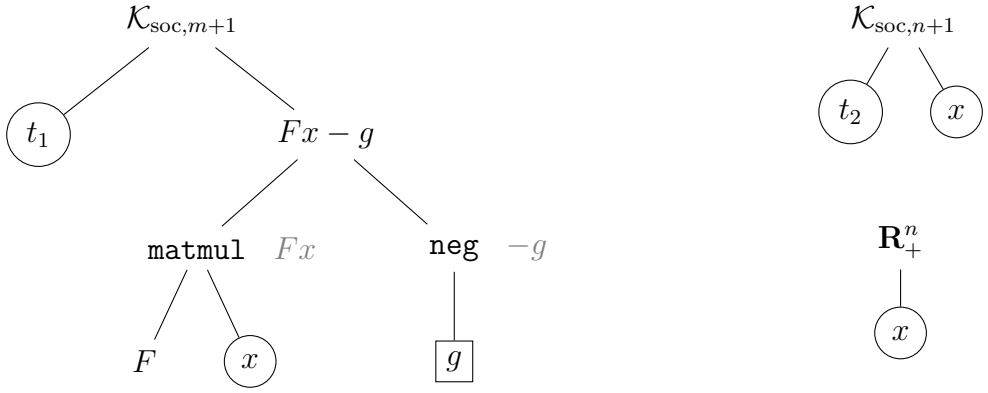
$$\begin{array}{ll} \text{minimize} & t_1 + \mu t_2 \\ \text{subject to} & x \in \mathbf{R}^n_+ \\ & (t_1, Fx - g) \in \mathcal{K}_{\text{soc},m+1} \\ & (t_2, x) \in \mathcal{K}_{\text{soc},n+1} \end{array} \tag{1.6}$$

where t_1 and t_2 are new epigraph variables, and the cone $\mathcal{K}$ from (1.2) is the product $\mathbf{R}^n_+ \times \mathcal{K}_{\text{soc},m+1} \times \mathcal{K}_{\text{soc},n+1}$. Problem (1.6) is now in the functional conic form of §1.3.1: every

constraint is a cone membership, and the objective and constraint expressions are affine. Comparing the DCP trees of Figure 1.2 with the trees of Figure 1.3 shows the effect of canonicalization: the nonlinear norm atoms have been replaced by epigraph variables, leaving affine expression trees in the variables, which we call the *conic-form trees* of the problem. The three rows of (1.6) remain three separate cone constraints, each carrying the affine expressions that must lie in its cone; Figure 1.3 accordingly shows one expression tree per constraint, rather than a single combined tree.



(a) Expression tree of the objective



(b) One expression tree per constraint

Figure 1.3 *After canonicalization.* `CVXPY` represents the conic form (1.6) as these expression trees: the nonlinear norm atoms have become epigraph variables, leaving the objective and the three separate cone constraints affine. The variable x appears as a leaf of several trees; all occurrences refer to the same variable.

1.4 Computing derivatives

Computing derivatives is a fundamental task in scientific computing, with many methods that differ in cost, in accuracy, and in the form of the result [40]. The derivatives of this thesis are large sparse matrices, the constraint Jacobian A and the objective Hessian P of a

canonicalized problem, so our focus is on producing *sparse* derivatives. Most differentiation software instead targets dense derivatives of scalar expression graphs, and this gap organizes both the present section and the survey of §2.2. This section presents the derivative objects, the elementary methods for computing them, and the choices that distinguish the software implementations; §2.2 develops the sparse case and the systems that compute it.

The Jacobian. For a vector-valued map $F : \mathbf{R}^n \to \mathbf{R}^m$ of an input $x \in \mathbf{R}^n$, the first-order derivative is the *Jacobian* $J \in \mathbf{R}^{m\times n}$ of partial derivatives of the outputs with respect to the inputs,

$$J_{ij} = \frac{\partial F_i}{\partial x_j}, \qquad i = 1, \ldots, m, \quad j = 1, \ldots, n. \tag{1.7}$$

We mainly consider the *constraint function*, which stacks the potentially many affine expressions of a canonicalized problem into a single vector function of the variables. Its Jacobian is the constant constraint matrix A of (1.2), whose sparsity pattern is fixed although its values may change from one solve to the next.

The Hessian. For a scalar function $f : \mathbf{R}^n \to \mathbf{R}$, the second-order derivative is the *Hessian* $H \in \mathbf{R}^{n\times n}$ of second partial derivatives,

$$H_{ij} = \frac{\partial^2 f}{\partial x_i \, \partial x_j}, \qquad i, j = 1, \ldots, n. \tag{1.8}$$

The Hessian is the Jacobian of the gradient of f, and is symmetric whenever f is twice continuously differentiable. It can therefore be obtained by differentiating the gradient with the same methods used for the Jacobian, although dedicated methods that compute it directly also exist. The function we consider is the canonicalized objective, whose Hessian is the matrix P of (1.2).

Numerical differentiation. Other than computing derivatives by hand, the simplest method is to use *numerical differentiation*, which approximates the derivative of a function at a point by evaluating the function at nearby points using a *finite difference* formula such as

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}, \tag{1.9}$$

where h is a small step size (the central variant $f'(x) \approx (f(x+h) - f(x-h))/2h$ improves accuracy at the cost of one more evaluation). The approach is simple but has two downsides. First, it requires a linear number of function evaluations with respect to the number of input variables, which is expensive for high-dimensional functions. Second, it can easily lead to

truncation errors (when h is too large) or *rounding errors* (when h is too small) in the computed derivative if the step size is not carefully chosen [41]. Despite these downsides, numerical differentiation is still widely used in practice, especially for verifying the correctness of derivative computations (also known as a *derivative checker*) [42].

Symbolic differentiation. Mathematical expressions can be manipulated automatically using simple derivative rules of calculus, such as the product and sum rule. This *symbolic* approach to differentiation is used in computer algebra systems such as `Mathematica` [43] and `Maple` [44] as well as the `SymPy` library in Python [45]. Unlike its numerical counterpart, symbolic differentiation tools compute derivatives as closed-form expressions, which can be simplified and evaluated at any arbitrary point. These closed-form expressions are particularly useful for deriving analytic solutions of optimization problems, and can be further used in mathematical proofs and analyses.

However, a naive implementation of symbolic differentiation can quickly form exponentially large symbolic expressions, which is impractical and inefficient for many problems [46]. When intermediate expressions are stored as *common subexpressions*, and the expression is represented with a *directed acyclic graph* (DAG) instead of a tree, symbolic differentiation performs the same set of operations as *reverse mode differentiation* [47].

Algorithmic differentiation. *Algorithmic differentiation* treats the function as a *computational graph* of elementary operations, each with a known local derivative rule. The chain rule is then applied to combine these local derivatives throughout the graph, giving the derivative of the whole function. AD is also commonly referred to as *automatic differentiation*; we prefer the term *algorithmic* to emphasize that AD differentiates the algorithm that computes a function, rather than a numerical approximation or a symbolic formula [46]. In this broad sense, AD can refer to any method that applies the chain rule over a computational graph. Its implementations differ in four basic choices: whether the graph is *scalar* or *vectorized*, whether the chain rule runs *forward* or in *reverse*, whether the derivative is assembled *directly* or accessed through an *oracle* of matrix–vector products, and whether the result is stored *sparse* or *dense*.

Scalar and vectorized graphs. The first choice is what a node of the graph represents. In most AD tools, every node is a single elementary operation, an addition, a multiplication, or a univariate function, and every edge carries a scalar partial derivative [46]. A *vectorized* graph coarsens this representation: a node is a whole vector or matrix operation, and its local derivative is a matrix, a block of the Jacobian. Nearly all tools operate on scalar graphs; the

vectorized exceptions include the MX expressions of `CasADi` [12] and JAX [48], although the latter still stores its derivatives densely.

Modes of differentiation. The chain rule can be applied in different orders, all giving the same result. Writing the derivative as a product of local derivatives, *forward mode* (or *tangent* mode) associates the product from the input side, following the order of evaluation, while *reverse mode* (or *adjoint* mode) associates it from the output side, which requires storing the intermediate values of the evaluation [40]. The two modes have asymmetric costs: in forward mode the work scales with the number of inputs, and in reverse mode with the number of outputs. The mode is thus chosen per derivative object rather than fixed by the tool: a gradient has a single output and calls for reverse mode, a Jacobian for whichever of its two dimensions is smaller, and most tools provide both modes.

Oracle and direct assembly. The third choice is what the differentiation returns. An *oracle* returns the action of the derivative on a vector, the product Jv in forward mode or $J^\top w$ in reverse mode, without ever forming J. One product costs one *sweep* of the graph: the function is evaluated once, and every node carries one extra scalar, its derivative along the seed direction v, updated by the chain rule as the evaluation proceeds. A sweep thus costs about one function evaluation, and the per-node state stays a single scalar; recovering J column by column takes one sweep per column, since propagating all n directions at once means every node carries a whole gradient rather than a scalar. Many applications need only the action of the derivative, as in training neural networks or in iterative linear solvers, but the conic solvers targeted in this thesis consume the matrix itself as problem data. A *direct* method accepts exactly that trade: it assembles J in a single sweep that carries whole derivative objects, dense or sparse, through the graph, each node combining the local derivatives of its arguments. The choice between the two matters only when the target matrix is sparse, which is the last of the four choices.

The sparse case. When the target matrix is sparse, the two assemblies part ways. With an oracle, the sparsity pattern must be detected and *graph coloring* used to compress the many sweeps into few. With direct assembly, the local derivatives are carried through the graph as sparse matrices and combined into the result, so there is nothing to compress. The four choices are not independent: direct assembly is practical only on vectorized graphs, whose few coarse nodes make carrying a matrix per node affordable. This thesis assembles the sparse Jacobian and Hessian directly on a vectorized graph, without oracle sweeps; §2.2 develops both assemblies and locates the systems of the literature, and our approach, among

them.

1.5 Matrix computations

Matrix computations appear at every stage of this thesis. The problem data P and A are sparse matrices, and matrix stuffing assembles them from smaller intermediate matrices. This section reviews the two families of matrix representations, dense and sparse, each from the standpoint of both storage and algorithms.

1.5.1 Dense matrices

Dense storage. A dense matrix is stored as a single contiguous array of its entries, ordered row by row (row-major) or column by column (column-major). The position of an entry is computed directly from its indices, so no index arrays are needed, and rows or columns are traversed sequentially in memory.

The BLAS. Dense linear algebra is served by a standardized interface of vector and matrix kernels, the *Basic Linear Algebra Subprograms* (BLAS). The interface is organized in three levels by the ratio of computation to data: level 1 covers vector operations such as dot products and scaled additions [49], level 2 covers matrix–vector products, and level 3 covers matrix–matrix products [50]. LAPACK, the library behind dense factorizations and linear system solving in most scientific software, is built on top of these kernels [51]. Because the interface is standard, decades of optimization effort concentrate behind it: implementations such as `OpenBLAS`, Intel `MKL`, and Apple `Accelerate` are tuned to each processor family, and every program calling the interface benefits. In practice, an optimized implementation is available on nearly every machine, whether shipped with the operating system or bundled by numerical libraries.

Why dense kernels are fast. The performance of optimized BLAS implementations comes from working with, rather than against, the memory system. A modern processor multiplies numbers far faster than it loads them from main memory, so fast code must reuse data already sitting in its caches and must read memory contiguously. Dense storage makes both possible: optimized kernels traverse arrays in memory order, process several entries per instruction with vector (SIMD) units, and tile matrix–matrix products into blocks that fit in cache [52]. Level-3 kernels benefit the most: they perform $O(n^3)$ operations on $O(n^2)$ data, so each entry loaded from memory is reused many times. These speeds, however, rely on every entry being stored and processed, zeros included.

1.5.2 Sparse matrices

In optimization problems, the matrices P and A contain mostly zero entries; such matrices are called *sparse*. The simplest sparse matrix data structure is the *coordinate* or *triplet* format. It consists of two integer index arrays `rows` and `cols` and a floating point value array `vals` of length equal to the number of *nonzero entries* in the matrix.

For example, the matrix

$$S = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 2 & 0 & 3 & 4 \\ 0 & 5 & 6 & 0 \end{bmatrix} \tag{1.10}$$

can be represented in coordinate format as

```
int    rows[] = {0, 1, 1,  1,  2,  2};
int    cols[] = {1, 0, 2,  3,  1,  2};
double vals[] = {1, 2, 3, 4,  5,  6};
```

where we use zero-based indexing and array declaration notation from the C programming language [53]. The coordinate format is the easiest to understand and form, but it is not effective for most operations on sparse matrices; the *compressed sparse row* or CSR format is commonly used instead.

Compressed sparse row format. The CSR format differs from the coordinate format in that it replaces the `rows` array with a row pointer array `idxptr` of length equal to the number of rows plus one. The row pointer array indicates the starting index of each row in the `cols` arrays. For example, the matrix in (1.10) can be represented in CSR format as

```
int  idxptr[] = {0, 1,         4,      6};
int    cols[] = {1, 0, 2,  3,  1,  2};
double vals[] = {1, 2, 3, 4,  5,  6};
```

where we deliberately format `idxptr` to align with the start of each row in the `cols` arrays. The benefits of the CSR format are more than a reduced size of the `idxptr` array compared to the `rows` array. In the CSR format, accessing a row takes constant time and is very *cache-friendly* since the nonzero entries are stored contiguously in memory. For this reason, the CSR format is the default for *iterative methods* to solve sparse linear systems [54].

Compressed sparse column format. The *compressed sparse column* or CSC format applies the same compression to the columns instead: `idxptr` now points into a `rows` array, and the nonzero entries are stored column by column. The CSC arrays of a matrix read

exactly as the CSR arrays of its transpose, so transposition is a conversion between the two formats, computable without sorting in one counting pass over the nonzeros [55]. Accessing a column of a CSR matrix instead requires a scan of every row, so algorithms that traverse columns prefer CSC; direct methods for sparse linear systems are the standard example [56].

Matrix-vector products. Both compressed formats multiply a dense vector in one pass over the nonzeros, but the loop must follow the storage: CSR *gathers*, computing each output entry as a sparse dot product along its row, while CSC *scatters*, accumulating each column's contribution into the output vector.

Matrix-matrix products. The product of two sparse matrices is harder than a matrix-vector product because the sparsity pattern of the result is not known in advance. The standard algorithm is due to Gustavson [55]: the result is formed row by row, each row as a combination of the rows of the second operand selected by the nonzeros of the first, gathered in a dense accumulator before compaction into sparse storage [57]. The algorithm underlies the sparse products of widely used linear algebra libraries such as `Eigen` [63] and `scipy.sparse`.

Other sparse formats. The coordinate, CSR and CSC formats are the most widely used sparse matrix formats, but many others exist for different specialized use cases. The scientific Python sparse library `scipy.sparse` [58] allows users to form other types of sparse matrices such as the list of lists (LIL) matrix, the dictionary of keys (DOK) matrix, and the block-sparse row (BSR) matrix.

CHAPTER 2 LITERATURE REVIEW

This chapter reviews the implementation of `CVXPY`'s matrix stuffing (§2.1) and of other modeling language approaches, from sparse AD (§2.2) to `AMPL` and `JuMP` (§2.3). These modeling languages have a similar purpose, and the goal of the chapter is to make the distinctions clear. The backends of `CVXPY` compute the same problem data from the same input as `SparseDiffEngine`, and they serve as the baselines for the benchmarks of Chapter 4. `CasADi` carries vectorized expressions, but must discover the structure of their derivatives at runtime, over many sweeps of the graph. `AMPL` and `JuMP` let the user declare vectorized expressions, and scalarize them for processing.

2.1 Matrix stuffing in `CVXPY`

`CVXPY` takes the perspective of matrix calculus [59] for computing the coefficient matrices of linear expressions, and this is where most of the computational burden of canonicalization lies. The section gives a brief summary of matrix calculus and a few examples, presents the conventions of `CVXPY`'s backend, and closes with implementation details that become important when contrasting it with our methodology (§3).

Matrix calculus. Matrix calculus extends differentiation to functions whose arguments and values are vectors and matrices. Systematic rules for such derivatives date back to the 1940s [60]. The simplest convention is to vectorize [59]: every expression is identified with

$$\mathbf{vec}(X) = \begin{bmatrix} x_{11} & \cdots & x_{m1} & x_{12} & \cdots & x_{mn} \end{bmatrix}^T, \tag{2.1}$$

the vector that stacks the columns of $X \in \mathbf{R}^{m \times n}$, and the derivative of one vectorized expression with respect to another is an ordinary Jacobian matrix. Under this convention matrix operations become linear operators on the vectorized arguments, and their Jacobians carry structure: for example

$$\mathbf{vec}(FX) = (I \otimes F)\ \mathbf{vec}(X), \tag{2.2}$$

so the matrix product FX corresponds to the Kronecker-structured, block-diagonal Jacobian $I \otimes F$. The convention is illustrated in Figure 2.1.

$$\texttt{X = cp.Variable((2, 2))} \quad \Longrightarrow \quad \frac{\partial\, \mathbf{vec}(X)}{\partial\, \mathbf{vec}(X)} = \begin{array}{c} \\ x_{11} \\ x_{21} \\ x_{12} \\ x_{22} \end{array} \begin{array}{c} \begin{array}{cccc} x_{11} & x_{21} & x_{12} & x_{22} \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

(a) A matrix variable is the identity map of its own entries.

$$\texttt{Y = F @ X} \quad \Longrightarrow \quad \frac{\partial\, \mathbf{vec}(Y)}{\partial\, \mathbf{vec}(X)} = I \otimes F = \begin{array}{c} \\ y_{11} \\ y_{21} \\ y_{12} \\ y_{22} \end{array} \begin{array}{c} \begin{array}{cccc} x_{11} & x_{21} & x_{12} & x_{22} \end{array} \\ \begin{bmatrix} f_{11} & f_{12} & 0 & 0 \\ f_{21} & f_{22} & 0 & 0 \\ 0 & 0 & f_{11} & f_{12} \\ 0 & 0 & f_{21} & f_{22} \end{bmatrix} \end{array}$$

(b) The Jacobian (2.2) of the product $Y = FX$ is block-diagonal, one copy of F per column of X.

Figure 2.1 The vec convention on a matrix variable X and the product FX, with $X, F \in \mathbf{R}^{2\times 2}$. The entries of $\mathbf{vec}(X)$, in column-major order, index the columns of the Jacobians.

Composing linear operators. After canonicalization every expression is affine, so each node of a conic-form tree can be written in the vectorized form

$$f(x) = A\,x + b = \begin{bmatrix} A & b \end{bmatrix} \begin{bmatrix} x \\ 1 \end{bmatrix}, \tag{2.3}$$

where x is the stacked vector of all the problem variables, A is the Jacobian of the node's expression f with respect to x, constant since f is affine, and b is a constant vector. Each variable occupies a contiguous block of columns of A, starting at its *column offset* within x, and the columns of the variables that do not appear under the node are zero. Matrix stuffing computes the augmented matrix $[A\ b]$ of every node in one bottom-up pass. A variable leaf contributes an identity block at its own column offset, and a constant leaf contributes its value to the last column. An interior atom g with arguments $f_1, \ldots, f_k$ is a linear map $g = \sum_i L_i f_i$, and since each L_i is a constant matrix under the vec convention, the chain rule reduces to one matrix product per child:

$$\begin{bmatrix} A & b \end{bmatrix} = \sum_{i=1}^{k} L_i \begin{bmatrix} A_i & b_i \end{bmatrix}. \tag{2.4}$$

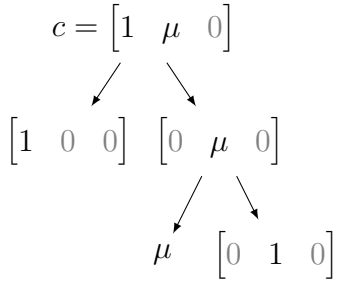
Computing the problem data thus amounts to composing linear operators over the tree, with no differentiation at the nodes. The node matrices of the figures in this section are the matrices $[A\ b]$, and the final result of the walk is the problem data.

Views and copies. The linear operators divide into *views*, which involve no atom data, and *copies*, which do. A view reuses the stored coefficients of its argument: negation flips the sign of each entry, indexing keeps a subset of the rows, and reshaping leaves the vectorization unchanged, so the cost of a view is at most proportional to the number of stored nonzeros. A copy occurs at the bivariate atoms that carry data. In a matrix product, an elementwise multiplication, or a division, one operand is constant under the DCP rules, and the operator multiplies a Kronecker or diagonal matrix formed from this constant operand with the augmented matrix of the other, storing the product as a new coefficient matrix with a new sparsity pattern. Table 2.1 classifies a subset of the operators, where $\mathcal{I}$ denotes an index set and $\mathbf{diag}(v)$ the diagonal matrix with the entries of v on its diagonal.

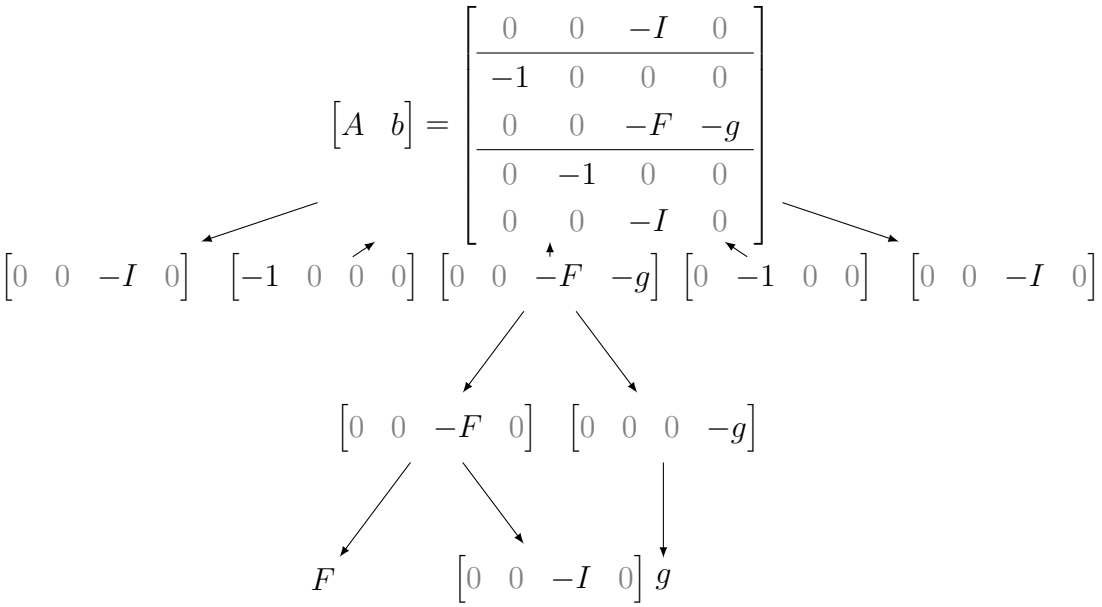
Table 2.1 A subset of the linear operators, classified as views or copies, applied to an argument f_1 with augmented matrix $[A_1\ b_1]$.

Atom	**Operator** L	**Action on** $[A_1\ b_1]$
Views		
$-f_1$	$-I$	flip the sign of every stored entry
f_1^T	P, the vec-permutation	permute the rows
$f_1[\mathcal{I}]$	$P_{\mathcal{I}}$, a row selection	keep the rows indexed by $\mathcal{I}$
reshape(f_1)	I	none, the vectorization is unchanged
Copies		
Ff_1	$I \otimes F$	multiply by a block diagonal formed from F
f_1B	$B^T \otimes I$	multiply by a Kronecker matrix formed from B
$c \odot f_1$	$\mathbf{diag}(\mathbf{vec}(c))$	multiply by a diagonal formed from c
f_1/c	$\mathbf{diag}(\mathbf{vec}(c))^{-1}$	multiply by the reciprocal diagonal

Example. The walk on the conic-form trees of problem (1.5) is traced in Figure 2.2: the leaves contribute their blocks and offsets, the interior nodes apply their operators, and the root stacks its children at their constraint row offsets into $[A\ b]$. The rest of the section describes how `CVXPY` implements this walk.



(a) Matrix stuffing of the objective



(b) Matrix stuffing tree of the constraints

Figure 2.2 Matrix stuffing of problem (1.5) by a recursive walk of the conic-form expression trees of Figure 1.3: each node's matrix is obtained by applying a linear operation to the matrices of its children. The horizontal rules in $[A\ b]$ separate the rows of the three constraints.

Local versus global derivatives. The augmented matrices of (2.3) can be laid out in two ways (Figure 2.3). `CVXPY` stores them *locally*: each node keeps one coefficient block per variable appearing beneath it, keyed by the variable's identifier, and every operator is applied to each block in turn. A block's columns index only the entries of its own variable, so the column offsets are not needed during the walk. The constant data is stored under the key -1 and becomes the last column of the assembled matrix. Only after the walk completes are the blocks shifted to their variables' column offsets and concatenated into the full matrix $[A\ b]$.

In the alternative, *global* layout, each node stores one matrix whose columns span the whole stacked vector x, and the column offsets are applied from the start rather than at the end. The local layout keeps the blocks small and skips the variables absent from a subtree, but every operator runs once per stored block. The engine of §3 uses the global layout.

$$\left\{ x \mapsto \begin{matrix} x \\ \begin{bmatrix} -F \end{bmatrix} \end{matrix},\ -1 \mapsto \begin{matrix} 1 \\ \begin{bmatrix} -g \end{bmatrix} \end{matrix} \right\} \qquad \begin{matrix} t_1 & t_2 & x & 1 \\ \begin{bmatrix} 0 & 0 & -F & -g \end{bmatrix} \end{matrix}$$

(a) Local layout. (b) Global layout.

Figure 2.3 What each layout stores during the walk, at the node $-(Fx-g)$ of problem (1.5): a dictionary of narrow per-variable blocks, or a single matrix spanning all columns.

Parameters and tensors. In a problem family the structure is fixed and only the numeric parameter values change between solves. Disciplined Parameterized Programming (DPP) restricts how parameters may enter such families: only affinely, so that the problem data is affine in the parameters [25]. `CVXPY` accounts for this with a third tensor axis. Every coefficient block stores one slice per parameter entry plus one slice for the constant part, and the operators of the walk apply to each slice. Flattening the slices of the root yields the cached *parameter map*

$$\begin{bmatrix} c \\ d \end{bmatrix} = Q\,\tilde{\theta}, \qquad \tilde{\theta} = \begin{bmatrix} \theta \\ 1 \end{bmatrix}, \tag{2.5}$$

shown in Figure 2.4 for the objective of problem (1.5); the constraint data $[A\ b]$ is recovered the same way from its own map. After a parameter change, a re-solve costs a single sparse matrix–vector product $Q\tilde{\theta}$.

$$Q = \begin{matrix} \\ t_1 \\ t_2 \\ x \\ d \end{matrix} \begin{matrix} \begin{matrix} g^T & \mu & 1 \end{matrix} \\ \begin{bmatrix} 0\cdots 0 & 0 & 1 \\ 0\cdots 0 & 1 & 0 \\ 0\cdots 0 & 0 & 0 \\ 0\cdots 0 & 0 & 0 \end{bmatrix} \end{matrix} \qquad \begin{bmatrix} c \\ d \end{bmatrix} = Q\,\tilde{\theta} = \begin{bmatrix} 1 \\ \mu \\ 0 \\ 0 \end{bmatrix}$$

Figure 2.4 The DPP path on the objective data of problem (1.5): the parameter map Q is built and cached once, and each re-solve recovers (c, d) with the single sparse product $Q\tilde{\theta}$.

The `ignore_dpp` path. Problems that fall outside the DPP grammar, or that opt out of it by solving with `ignore_dpp=True`, take a simpler path. The reduction `EvalParams` replaces each parameter leaf by a constant holding its current value, and the resulting parameter-free tree goes through the walk as if the problem had no parameters (Figure 2.5). The result is the same objective data c as in Figure 2.4, now obtained by re-walking the tree instead of applying the cached product $Q\tilde{\theta}$. The path is fully general, since arbitrary functions of parameters reduce to constants, but every change of θ repeats the whole symbolic walk even though the sparsity pattern of the problem data never changes.

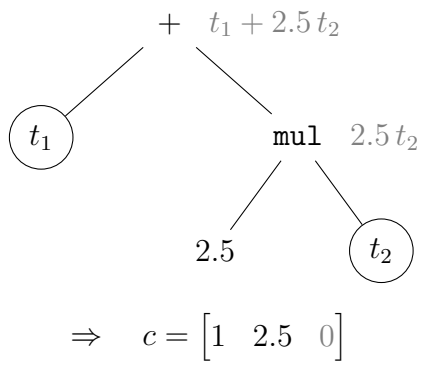


Figure 2.5 The `ignore_dpp` path on the objective of problem (1.5): the parameter leaf μ is replaced by a constant holding its current value 2.5, and the parameter-free tree is re-walked on every change of the parameters.

The three backends. `CVXPY` implements the walk in three canonicalization backends that differ in how they represent the parameter axis. The C++ backend `cvxcore` keeps one sparse matrix per parameter slice and loops over the slices at every operator. The `SciPy` backend stacks the slices vertically into a single two-dimensional sparse matrix. The `COO` backend does not form the tensor; it stores each nonzero with an explicit parameter index alongside its row and column.

Code generation. When canonicalization remains too slow, `CVXPYgen` can compile the problem ahead of time: it emits problem-specific C code that embeds the cached parameter map (2.5) of a DPP problem [61], continuing an earlier line of work on code generation for embedded second-order cone programs [62]. The benchmarks of §4 do not compare against it: code generation is an offline step that fixes the problem structure at compile time, a different deployment model from a runtime backend that must accept arbitrary problems.

Implementation details. Two implementation details, shared by all the backends, limit the current approaches in `CVXPY`.

- *External sparse libraries.* The backends delegate their matrix operations to a general-purpose sparse library, Eigen [63] in `cvxcore` and `scipy.sparse` [58] in the Python backends. Although the COO backend stores no `scipy.sparse` matrices, it still converts to them for sparse operations such as matrix products and Kronecker products. The storage formats and kernels are fixed by the library, leaving no room to exploit structure known at canonicalization time. In `cvxcore` the views of Table 2.1 also materialize explicit sparse operators: a reshape builds a sparse identity matrix that is then multiplied.
- *Everything is sparse.* All coefficient data is treated as sparse: dense atom data is converted to a sparse matrix on entry, and no operator carries a dense or block special case, so dense blocks pay sparse-format overhead and no BLAS kernel (§1.5.1) is used.

Other design choices matter when comparing with `SparseDiffEngine`: each node stores a dictionary of local Jacobian blocks rather than a single global matrix (Figure 2.3), and the quadratic forms of the objective are removed by substitution and recovered in a separate matrix stuffing pass that assembles the objective matrix P. In §3 we revisit these limitations and design choices, which motivate our implementation.

References. Michael Grant's papers and PhD thesis are the standard references on verifying convexity and canonicalization [15, 16, 33]. The implementation details of this section come from the `CVXPY` source code [64].

2.2 Sparse derivatives from detected structure

The introduction (§1.4) organized differentiation software by four choices: the granularity of the expression graph, the mode of differentiation, the assembly of the result, and its storage. This section examines the assembly choice in detail, with emphasis on oracle assembly, the standard of the literature [46]. Under oracle assembly, sparsity detection and graph coloring act as compression: they reduce the number of oracle sweeps needed to recover the sparse matrix. Table 2.2 gathers recent systems, and the section closes with the system nearest this thesis, `CasADi`.

Scalar traces. Early AD tools such as ADOL-C [65] and CppAD [66] record the scalar graph as a runtime trace of every elementary operation [46,67]. Such a trace can be enormous:

a single matrix–vector product Ax unrolls into one scalar multiplication for every entry of A, together with the additions that sum them, so it quickly reaches millions of nodes for a realistic problem.

Oracle-based differentiation. The oracle of §1.4 returns the Jacobian-vector product (JVP) Jv in forward mode and the vector-Jacobian product (VJP) $J^\top w$ in reverse mode (for the Hessian, the analogous Hessian-vector product Hv comes from one forward sweep over one reverse sweep) [40]. These matrix-vector products are obtained without ever forming J, each by one sweep of the graph at about the cost of one function evaluation; borrowing terminology from numerical linear algebra, this is also called the *matrix-free* view of AD. Figure 2.6 traces one forward sweep on a small graph.

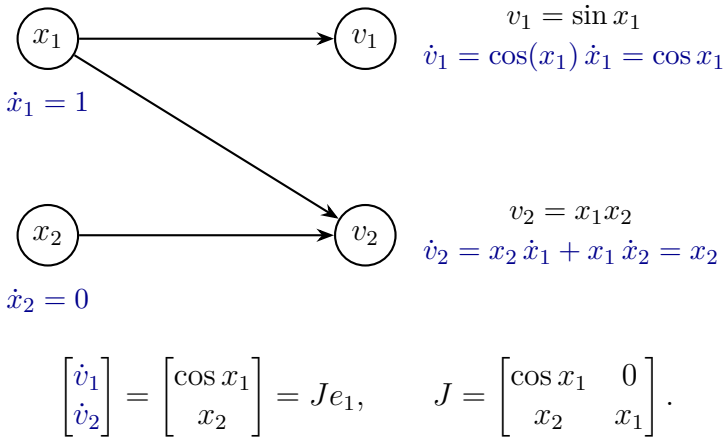


Figure 2.6 One forward sweep of the computational graph of $F(x_1, x_2) = (\sin x_1,\ x_1 x_2)$. Alongside its value, each node carries a single extra scalar, its tangent $\dot{v}$ (in color), updated by the chain rule as the evaluation proceeds. The seed $(\dot{x}_1, \dot{x}_2) = (1, 0)$ delivers the first column of J at the outputs; every further column requires its own sweep.

Recovering the Jacobian. Recovering J explicitly requires *sweeping* the oracle over a sequence of seed vectors: the n unit seeds recover it column by column in forward mode, and the m unit seeds recover it row by row in reverse mode (the Hessian is recovered the same way, though its symmetry permits fewer seeds) [46]. Seeding the full identity recovers J exactly, $JI = J$, so assembling a dense J is simply the oracle batched over the unit seeds. When the Jacobian is sparse, most of each sweep is wasted on zeros, and the full identity is prohibitively expensive; reducing the number of sweeps is the purpose of graph coloring. Coloring operates on the oracle alone, independently of how the underlying graph is built, so even fully vectorized frameworks such as JAX [48], whose local derivatives are Jacobian

blocks combined by the rules of matrix calculus [59], rely on it when they need an assembled sparse Jacobian.

Graph coloring. The number of sweeps drops when several seeds share one. Two columns of J with no nonzero in a common row are *structurally orthogonal*, so one seed recovers both without their nonzeros colliding. Finding the fewest such groups is a graph-coloring problem on the column intersection graph, whose edges join columns that share a row [26, 27, 68]. Figure 2.7 illustrates the compression: one sweep per color group replaces one sweep per column. The Hessian compresses further: its symmetry admits dedicated star and acyclic colorings that use fewer colors than the general Jacobian case [69].

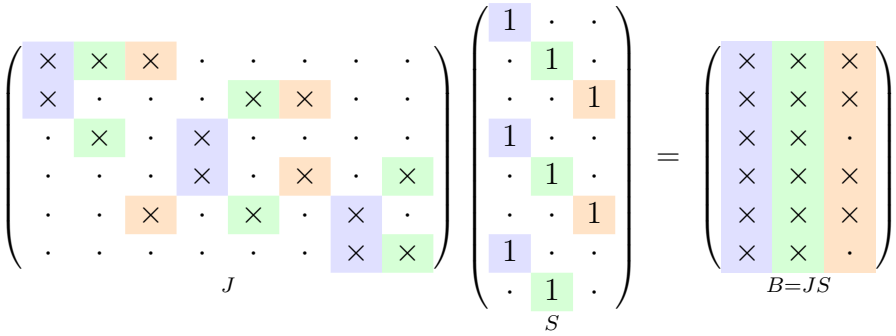


Figure 2.7 Graph coloring compresses the sweeps of the oracle. The columns of the sparse Jacobian J are grouped into three structurally orthogonal color classes; the seed matrix S sums the unit vectors of each class, and the compressed product $B = JS$ recovers every nonzero of J without collisions, in three sweeps instead of eight.

Detect, compress, propagate. An oracle-based sparse Jacobian is obtained in three stages: *detect* the sparsity pattern, *compress* it by coloring, and *propagate* the compressed seeds of Figure 2.7 through the oracle to recover the nonzero entries [70]. The pattern is computed once and reused, so its cost is amortized across repeated evaluations of the same function. This scheme underlies sparse AD in oracle-based tools such as `CasADi` (§2.2.1).

Direct differentiation. Direct assembly (§1.4) needs neither sweeps nor coloring: each node carries its whole derivative object, and one pass over the graph combines them into J. On a scalar graph the cost is prohibitive. The derivative work at a node is proportional to the gradient it carries, not to the $O(1)$ operation it performs, and every intermediate materializes a gradient of its own: the inner product $y = a_1x_1 + \cdots + a_kx_k$ unrolls into $k - 1$ running

partial sums whose gradients grow from one nonzero to k, so direct propagation costs $O(k^2)$ against $O(k)$ for one sweep. The affine expressions of canonicalized problems are dominated by such sums. Direct assembly on scalar graphs is therefore rare: *vertex elimination* (also called *cross-country* accumulation) forms J by eliminating the intermediate vertices of the linearized graph, but the optimal elimination order is intractable [46, 71], and the scalar symbolic graphs of `CasADi` build derivative expressions node by node in the same spirit [72]. Vectorization removes the obstruction: a node is a whole matrix operation, its local derivative is a structured block rather than a 0/1 dependency, and the graph shrinks from millions of scalar operations to a few structured nodes, so carrying a sparse matrix per node is affordable. Recent works apply this idea in statistics [73] and optimal control [74]. On a vectorized graph, direct propagation is cheaper when the local blocks are structured and intermediate fill-in stays bounded; the oracle with coloring is cheaper when the intermediates densify. Which assembly is preferable follows the solver that consumes the result: the conic solvers targeted in this thesis require the matrix explicitly as data, which the direct approach produces.

Table 2.2 Differentiation methods classified by the four choices of §1.4: the *expression type* (scalar or vectorized), the *mode* (forward, reverse, or both), the *assembly* of the Jacobian (formed *directly* by composing local Jacobians, or matrix-free through an *oracle* that returns only matrix–vector products and is reconstructed by seeded sweeps), and the *data format* (sparse or dense). The vectorized, forward, direct, sparse methods (lower block) need no graph coloring, unlike the matrix-free methods above them. The choices are not independent: every direct method in the table operates on a vectorized graph.

Method / work	**Expression type**	**Mode**	**Assembly**	**Data format**
ADOL-C, CppAD [65, 66]	scalar	both	oracle	dense
`CasADi` [12]	scalar / vectorized	both	oracle	sparse
JAX [48]	vectorized	both	oracle	dense
`torch-sla` [75]	vectorized	reverse	oracle	sparse
Hill and Dalle [70]	scalar	both	oracle	sparse
Zou and Jiang [74]	vectorized	forward	direct	sparse
Kwok et al. [73]	vectorized	forward	direct	sparse
∇SD [76]	vectorized	forward	direct	sparse
This thesis	**vectorized**	**forward**	**direct**	**sparse**

2.2.1 `CasADi`

`CasADi` [12] is an integrated symbolic-AD framework for optimization whose quadratic-programming interface extracts the explicit matrices P and A by AD on the expression graph; it is the reference implementation of the detect–compress–propagate scheme of this section in optimization software.

Expression types. Expressions come in two types [72]. An `SX` graph is scalar, with exact node-by-node sparsity, at the cost of a graph whose size grows with the problem data. An `MX` graph is vectorized: nodes are whole matrix operations, so large models keep small graphs. Either graph is compiled into a flat instruction tape over a shared work vector, which enables common-subexpression elimination, constant folding, and C code generation.

Jacobian assembly. Even on a vectorized graph, the Jacobian is reached through the oracle: a sweep over the tape carries a batch of seed directions and returns their products Jv, not J itself. Assembly therefore follows the stages of this section: the sparsity is computed by a pattern propagation over the tape, compressed by greedy distance-2 coloring, and the compressed seeds are pushed through batched forward or reverse sweeps, with the mode chosen by a cost heuristic on the pattern's shape. The Hessian is assembled as the Jacobian of the gradient, one forward sweep over the reverse tape, with a star coloring. Pattern, coloring, and the compiled derivative function are cached across evaluations. This is how `CasADi`'s own `qpsol` interface recovers the problem data of a quadratic program, by differentiating the model without invoking a solver:

```
import casadi as cs
z, th = cs.MX.sym("z", n), cs.MX.sym("th", p)
f, g = objective(z, th), slacks(z, th) # lowered conic model
c = cs.substitute(cs.gradient(f, z), z, 0)
P = cs.jacobian(cs.gradient(f, z), z) # Jacobian of the gradient
A = -cs.jacobian(g, z)
b = cs.substitute(g, z, 0)
data = cs.Function("data", [th], [P, c, A, b])
```

The dense-block worst case. A dense $m \times n$ block in the problem data makes every pair of its columns collide in some row: the column intersection graph restricted to the block is a clique, no coloring can use fewer than $\min(m, n)$ colors, and the sweep count grows to the full dimension of the block, even when the assembled matrix is mostly sparse. Lowered conic problems routinely embed dense blocks, so this worst case governs coefficient extraction in the setting of this thesis.

References. The book by Griewank and Walther [46] covers the theory and practice of scalar AD, including the graph-elimination view of Jacobian accumulation; the book by Naumann [67] focuses on implementation by operator overloading and source transformation. The survey by Baydin et al. [40] covers AD in machine learning. For the combinatorial side,

the survey of Gebremedhin et al. [27] is the standard entry point, and the illustrated guide of Hill and Dalle [77] is a modern tutorial on the detection–coloring–compression scheme. The `CasADi` paper [12] covers its expression graphs, AD implementation, and solver interfaces.

2.3 Coefficient extraction from declared structure

The systems of the previous section detect the structure of the problem at runtime and recover its coefficients through coloring. `AMPL` and `JuMP` need no detection: the structure is declared up front and the coefficients are stored as the user types them, so extraction reduces to reading off what is already known. This section examines both systems.

2.3.1 `AMPL`

`AMPL` [5] is the oldest system in this review, and its architecture still defines how most solvers receive nonlinear problems. The AMPL translator runs in its own process: it parses the model and the data, instantiates every constraint, and writes the resulting instance to a `.nl` file [28], which the solver, a separate executable linking the AMPL Solver Library (ASL), reads back [29].

The `.nl` format. The `.nl` file is an instance format, not a model format: sets and indexing are resolved by the translator, and what remains is one flat record per objective and constraint. A header states the dimensions of the problem and the nonzero counts of its Jacobian and gradients; the body is a sequence of segments keyed by a leading letter. `C` and `O` segments hold the nonlinear parts of constraints and objectives as expression graphs in Polish prefix notation; `J` and `G` segments hold their linear parts as explicit lists of (variable, coefficient) pairs; the remaining segments hold bounds and nonzero counts.

Example. The following quadratic program shows the syntax of `.nl` files.

```
var x; var y;
minimize obj: x^2 + x*y + 3*x;
subject to c1: x + 2*y >= 1;
```

The translator writes the following `.nl` file (ASCII encoding; annotations and elisions ours):

```
g3 0 1 0    # text format; 9 more header lines follow
 2 1 1 0 0  # 2 vars, 1 constraint, 1 objective, 0 ranges, 0 eqns
...
r           # constraint bounds: c1 >= 1
```

```
2 1
...
O0 0        # minimize objective 0; nonlinear part x^2 + x*y
o0          #   add
o5          #     pow
v0          #       variable 0 (x)
n2          #       constant 2
...         #     multiply subtree of x*y
J0 2        # linear Jacobian row of c1: two entries
0 1         #   coefficient 1 on variable 0
1 2         #   coefficient 2 on variable 1
G0 2        # linear part of the objective gradient: 3x
0 3
1 0
```

The quadratic terms $x^2 + xy$ appear only as the expression graph of the `O0` segment; every linear coefficient is spelled out in the `J` and `G` segments. For a linear program, the file *is* the problem data.

Solver-side evaluation. Inside the solver, ASL reads the linear coefficients and bounds directly from the file, so for a linear or quadratic-over-linear instance this read is the whole extraction, at a cost linear in the number of nonzeros. Only the rebuilt nonlinear graphs are differentiated, by reverse-mode AD [78], with sparse Hessians assembled from partial separability [79].

Advantages of the file format. The format has three benefits: any system that writes `.nl` reaches every solver that links ASL; the format has stayed backward compatible since 1990; and an instance is an ordinary file that can be archived, shipped to reproduce a bug, and rerun across machines. The cost is the decoupling itself: every solve serializes the instance to disk and rebuilds its expression graphs in the solver process.

2.3.2 `JuMP`

`JuMP` [11, 80] is an algebraic modeling language embedded in Julia; its abstraction layer between models and solvers is MOI [30]. Where `AMPL` decouples modeler and solver through a file, `JuMP` decouples them through a typed in-memory interface. For the linear, quadratic, and conic problems considered here, coefficients are stored as the user types them, and no differentiation occurs anywhere on the problem-data path.

Macros and term lists. The macros of `JuMP` rewrite each expression as it is entered: `@objective` and `@constraint` build affine and quadratic expression objects term by term, merging duplicates eagerly, so the model never holds an expression tree, only coefficient lists. These lists are typed by MOI. A `VectorAffineFunction` is a list of (row, variable, coefficient) terms plus a constant vector; a `ScalarQuadraticFunction` adds a list of (variable, variable, coefficient) triples. A constraint pairs one such function with a set, for instance a nonnegative orthant or a second-order cone: the functional conic form of §1.3.1. The constraint of the `AMPL` example above, entered as `@constraint(model, x + 2y >= 1)`, is stored as the pair

```
ScalarAffineFunction([(1.0, x), (2.0, y)], 0.0) in GreaterThan(1.0)
```

a term list, a constant, and a set; the expression itself is gone once the macro returns.

Extraction. Extraction is serialization: when the model is passed to a solver, MOI copies each constraint into the solver's own representation (`copy_to`); solvers that consume matrix data use the `MatrixOfConstraints` utility, which concatenates the stored term lists into one sparse matrix in the solver's format. The cost of extraction is this copy: a pure data-structure transformation, linear in the number of terms, with no analysis of any graph. The assembled matrix is the same problem data that `CVXPY`'s matrix stuffing computes (§2.1); `JuMP` arrives at it by copying stored coefficients rather than by differentiating conic-form trees.

Bridges. When a solver does not support a constraint type natively, MOI reformulates it through its *bridge* system: a bridge is a structural rewrite from one function-set pair to another, for instance from a rotated to a plain second-order cone, chosen by a shortest-path search over the graph of available bridges [30]. Bridging manipulates stored coefficients and adds auxiliary variables and constraints. This mirrors `CVXPY`'s canonicalization (§1.3.2) with one difference: `JuMP` rewrites problems whose data is already in matrix form, while `CVXPY` must first produce that data from a DCP atom tree, which is the work this thesis accelerates.

`JuMP` also has a nonlinear path, in which expression graphs are retained and derivatives come from sparse AD in the style of §2.2; it serves general nonlinear solvers and is out of scope for this thesis.

References. The MathOptInterface paper [30] is the best reference on the design of `JuMP`'s solver abstraction layer and its bridge system, and the MOI and `JuMP` documentation [81, 82] cover the implementation details. David Gay's technical reports [28, 29] remain the primary description of the `.nl` format and the `AMPL` solver interface.

Table 2.3 The reviewed systems and the engine of §3, sorted by the origin of their problem-data coefficients: stored as the user typed them, detected at runtime, or differentiated out of an expression graph.

System	Coefficients	Sparsity known from	Extraction cost
`JuMP`	stored	the term lists themselves	one copy of the term lists (`copy_to`)
`AMPL`	stored[a]	partial separability of the scalar tape	`.nl` write and ASL read
Hill and Dalle [70]	detected	index-set tracing, then coloring	compressed oracle sweeps
`CasADi` [12]	detected	pattern propagation, then coloring	compressed oracle sweeps
`CVXPY` [19]	differentiated	emerges from generic sparse products	operator walk over the conic-form trees
This thesis	**differentiated**	**analytic, per DCP atom**	**operator walk over the conic-form trees**

[a] Linear parts only; the nonlinear parts are differentiated by ASL in the solver process.

CHAPTER 3 A LIGHTWEIGHT CANONICALIZATION ENGINE

Mathematically, `SparseDiffEngine` computes the same derivatives as `CVXPY`'s canonicalization backend: both apply the chain rule of matrix calculus to trees of vectorized expressions, producing the Jacobian and Hessian of §1.4. The differences are in the implementation details, which we will present in this chapter. Firstly, the use of a custom sparse matrix data structure allows a natural split between the symbolic and numeric phases of the derivative computation. This separation is particularly useful for compiling problems involving parameters, where the structure is fixed but the numerical values can change. Secondly, in addition to being sparse, the data of most optimization problems also have a lot of structure, such as dense blocks or diagonality. In `SparseDiffEngine` we introduce two new matrix classes: `permuted-dense` and `stacked-permuted-dense` for this purpose; they enable us to make use of dense storage and BLAS subroutines when possible.

Finally, some important design decisions differ from `CVXPY`'s backends for simplicity rather than speed. Every node carries a single Jacobian whose columns span all of the variables of the problem, rather than a dictionary of local Jacobians keyed by variables. As a consequence, the chain rule is a single matrix product, and the constraint matrix is built by stacking the Jacobians of the constraints. In addition, quadratic functions pass through dedicated Hessian oracles instead of a separate substitution and matrix stuffing pass.

The rest of the chapter is structured as follows: we begin with a brief overview of the engine, describing how expressions are represented and how the code is organized. We then present the matrix representations used for the derivative computations, where we will focus much of our attention on the `permuted-dense` matrix, which appears most often. Next we describe the algorithms, or *kernels*, which allow us to operate on and combine these matrix representations. Rather than attempting to explain too much, we focus on the organization of the code and on how we handle operations between different types of matrices. An intrigued reader will learn more from exploring the repository, which is open source. The chapter ends with some comments about the Application Programming Interface (API) of `SparseDiffEngine` as well as how it is used within `CVXPY`.

3.1 Overview of the engine

`SparseDiffEngine` is a small C library which is around seventeen thousand lines of code (and a similar number of lines for the test suite). The expression node is defined in `expr.h`;

atoms that carry extra data, such as the exponent of a power, define a larger structure in `subexpr.h`. Each atom is implemented in its own file in the `atoms/` directory; there are around forty-five of them, divided into six subdirectories based on their mathematical properties: `affine`, elementwise and bivariate functions with full or restricted domains, and a small group of `other` atoms such as the quadratic form. The matrix data structures and the computational kernels that operate on them are in `utils/`, and `problem.c` collects the objective and constraints of a problem and implements the oracles that are passed to a solver.

Atom classes. Atoms in the same subdirectory compute their derivatives in the same way (Table 3.1). In this chapter we only discuss the affine atoms and the quadratic form, since these are the only atoms that appear in canonicalized DCP problems; the remaining classes are used by the nonlinear programming (NLP) interface of `CVXPY` [37]. Each atom

Table 3.1 The atom classes and how each computes its Jacobian.

Class	Examples	Jacobian computation
affine	`sum`, `index`, `left_matmul`	fixed; index maps and copies
elementwise, full domain	`exp`, `power`, `logistic`	diagonal scaling of the child Jacobian
elementwise, restricted domain	`log`, `entr`, `atanh`	diagonal at the variable's columns
bivariate, full domain	`multiply`, `matmul`	two-term chain rule
bivariate, restricted domain	`quad_over_lin`, `rel_entr`	two-term chain rule
other	`quad_form`, `prod`	specific to each atom

implements three operations: a *forward* pass, which evaluates the atom at a point; a *symbolic* pass, which computes the sparsity patterns of its derivatives and allocates memory for them; and a *numeric* pass, which computes the numerical values inside the fixed patterns. The constructor of the atom stores pointers to these three functions in the expression node.

The expression node. An expression node stores its dimensions, pointers to at most two children, a buffer for its value, and its derivative matrices, whose `n_vars` columns span all of the variables of the problem. A simplified version of the definition in `expr.h` is shown below.

```
typedef struct expr
{
    int d1, d2, size, n_vars, refcount, var_id;
    struct expr *left;
```

```
    struct expr *right;

    double *value;   /* forward value             */
    matrix *jacobian; /* Jacobian w.r.t. the variables */

    forward_fn      forward;
    jacobian_init_fn jacobian_init_impl; /* symbolic pass */
    eval_jacobian_fn eval_jacobian;    /* numeric pass */
    /* ... Hessian oracle, workspace and caching fields ... */
} expr;
```

Trees are built bottom up, with one constructor function per atom; the Python bindings expose the same constructors, so a `CVXPY` expression is converted by a single traversal that calls the constructor of the corresponding atom at each node (§3.4). Expressions that appear multiple times can be shared, in which case the tree becomes a directed acyclic graph. Nodes are therefore reference counted, and the symbolic pass returns early if a node was already initialized through another parent.

Leaf nodes. There are two leaf nodes, variables and parameters. A *variable* occupies a contiguous slice of the global variable vector, starting at its offset `var_id`; its Jacobian is a selection matrix with one unit entry per row, written once during the symbolic pass. Unlike `CVXPY`, there is no separate constant node: constants and parameters share a single leaf type, distinguished by an identifier that is -1 for constants and, for updatable parameters, the offset of the parameter in the global parameter vector. The Jacobian of this leaf is empty, since parameters are fixed during a solve, and because parameters are leaves of the tree, the derivative structure remains valid when their values change (§1.1).

3.2 Matrix representations

The `jacobian` field of the expression node has a polymorphic type called `matrix`, with one representation for each kind of structure that we want to exploit. Each atom chooses the representation of its Jacobian when it allocates it during the symbolic pass; multiplication by dense data allocates a permuted dense matrix, for example. After that, every operation on the matrix, such as transposition, row indexing, or a chain-rule product, simply dispatches to a kernel specific to the representations involved. The default representation is a generic sparse matrix stored in the CSR format of §1.5.2.

3.2.1 The permuted dense matrix

The Jacobians that appear during canonicalization are usually not sparse in the sense of having scattered, isolated nonzeros. Multiplication by problem data places the data itself, or a product involving it, as a dense block at a subset of the rows and columns of the Jacobian. Consider the expression Bx with dense data $B \in \mathbf{R}^{50\times 50}$, in a problem with ten thousand variables. Its Jacobian with respect to the full variable vector is a matrix with ten thousand columns that contains B in the fifty columns of x and is zero everywhere else. If this Jacobian is stored in a generic sparse format, every subsequent product goes through sparse kernels, and the hardware efficiency of §1.5.1 is lost.

Definition. We therefore store such matrices in *factored form*: we keep the dense block and the index vectors that place it, and never assemble the global matrix. A *permuted dense* matrix is a triple (X, r, c), where $X \in \mathbf{R}^{m_0\times n_0}$ is a dense block and r and c are strictly increasing index vectors, representing

$$A = S_r^T X S_c \in \mathbf{R}^{m\times n}, \qquad A_{r_i, c_j} = X_{ij}, \tag{3.1}$$

where $S_r \in \{0,1\}^{m_0\times m}$ and $S_c \in \{0,1\}^{n_0\times n}$ are the selector matrices whose rows are the rows of the identity indexed by r and c: multiplying a vector by S_r selects the entries indexed by r, and multiplying by S_r^T scatters them back into a vector of length m, with zeros elsewhere. All other entries of A are structurally zero and never stored. Figure 3.1 shows an example with a 3×3 block inside a 6×7 matrix.

$$A = \begin{pmatrix} \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ x_{00} & \cdot & x_{01} & \cdot & \cdot & x_{02} & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ x_{10} & \cdot & x_{11} & \cdot & \cdot & x_{12} & \cdot \\ x_{20} & \cdot & x_{21} & \cdot & \cdot & x_{22} & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \end{pmatrix} \qquad X = \begin{pmatrix} x_{00} & x_{01} & x_{02} \\ x_{10} & x_{11} & x_{12} \\ x_{20} & x_{21} & x_{22} \end{pmatrix}$$

$$r = (1, 3, 4), \qquad c = (0, 2, 5).$$

Figure 3.1 A permuted dense matrix. The dense block X is stored once, contiguously; the index vectors r and c place its entries in the global index space, and every entry outside the shaded cells is structurally zero and never stored. Each block row keeps its own shade, reused in Figure 3.2.

Storage. The block X is stored contiguously in row-major order and the index vectors as two integer arrays, so storage does not depend on the global dimensions. Two inverse maps, which send a global index to its block index or to -1, make membership checks constant time. Structure is computed with integer arithmetic on the index vectors; floating point operations run only on the contiguous block.

Structural operations. Every operation we need maps a permuted dense matrix to another permuted dense matrix. Transposition swaps the two index vectors and transposes the block. The structural atoms, such as indexing, broadcasting, and the row partitions of `sum`, work the same way: since row r_i of A is row i of X and every other row is zero, an operation on the rows of A reduces to the same operation on the rows of X, guided by the inverse row map.

output row i	0	1	2
requested global row s_i	3	0	4
inverse row map $r^{-1}[s_i]$	1	-1	2
action	copy $X_{1,:}$	zero row	copy $X_{2,:}$

$$C = A[(3,0,4),\,:] \;=\; \begin{pmatrix} x_{10} & \cdot & x_{11} & \cdot & \cdot & x_{12} & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ x_{20} & \cdot & x_{21} & \cdot & \cdot & x_{22} & \cdot \end{pmatrix} \begin{matrix} \leftarrow A_{3,:} \\ \leftarrow A_{0,:} = 0 \\ \leftarrow A_{4,:} \end{matrix}$$

$$X_C \;=\; X[(1,2),\,:] \;=\; \begin{pmatrix} x_{10} & x_{11} & x_{12} \\ x_{20} & x_{21} & x_{22} \end{pmatrix}, \qquad r_C = (0,2), \qquad c_C = c = (0,2,5).$$

Figure 3.2 Row indexing of the permuted dense matrix A of Figure 3.1. The table traces the scan of the inverse row map: each requested global row yields a block row of X to copy, or -1, in which case the output row is structurally zero and not stored. Shades follow the block rows of Figure 3.1. The result is the permuted dense triple (X_C, r_C, c_C).

Figure 3.2 traces the indexing of rows $(3,0,4)$ of the matrix in Figure 3.1: each hit in the inverse row map yields a block row to copy, and the miss at row 0 produces a zero row that is not stored. The result is again permuted dense, with block $X[(1,2),\,:]$, row vector $(0,2)$, and the column vector of A; a repeated row would simply copy the same block row twice. The structure costs one integer scan, the values two row copies.

The chain-rule product. The product of two permuted dense matrices is also permuted dense: for permuted dense A and B, the chain-rule product B^TA factors as

$$B^TA = S_{c_B}^T X_B^T \underbrace{\left(S_{r_B} S_{r_A}^T\right)}_{\text{row-set intersection}} X_A S_{c_A}, \tag{3.2}$$

The middle factor is a zero-one matrix whose entry (k, l) is one exactly when $r_{B,k} = r_{A,l}$, so the sparsity analysis of the product amounts to intersecting two sorted integer arrays. The result has row vector c_B and column vector c_A, which are known before any numerical values exist. For the values there are three cases. If the two row sets are equal, the product is computed on the two blocks with a single call to `dgemm`, the double precision general matrix–matrix product of level-3 BLAS (§1.5.1). If they overlap partially, the matching rows are first gathered into a workspace (allocated during the symbolic pass), followed by the same `dgemm`. If they are disjoint, the product is zero, and nothing is computed in the numeric pass. The weighted product B^TDA, needed for the Hessian of the objective, is handled the same way.

Performance. The main benefit of the factored form (3.1) is that the floating point work is done by dense kernels. If the same matrix is stored in the CSR format, its products go through sparse kernels, whose indexed accesses and irregular inner loops reach only a fraction of the throughput of BLAS [52] at the same operation count. In the factored form, the numeric pass of each kernel consists of a few BLAS calls on contiguous buffers. The product A^TDA, for example, is computed with one copy, m_0 scalings, and one `dgemm`, and the cost does not depend on the global dimensions.

Limitations. A permuted dense matrix cannot represent nonzeros outside its block: by (3.1), every such entry is structurally zero. A Jacobian that mixes a dense block with scattered entries simply uses the default CSR representation, and an operation whose result does not keep the block structure, such as adding operands with different patterns, is performed on CSR views of its operands. The remaining risk is dense data that contains many zeros, which would be stored and multiplied densely, spending floating point work on zeros; since `CVXPY` supplies sparse data in sparse formats, this does not arise in practice.

3.2.2 The stacked permuted dense matrix

When an expression with matrix shape $n \times p$ is multiplied by data $A \in \mathbf{R}^{m \times n}$, the effect on the vectorized expression is that of the block diagonal operator $I_p \otimes A$; this is the `left_matmul` atom. The operator is never formed: the engine keeps one copy of A and applies it block by

block. In memory, the p segments of length n of a vector x form a $p \times n$ row-major matrix $\tilde{X}$, and by a standard vectorization identity [59],

$$y = (I_p \otimes A)\, x \quad \Longleftrightarrow \quad \tilde{Y} = \tilde{X} A^T, \tag{3.3}$$

so applying the operator to a vector is a single `dgemm` on the stored A. Avoiding the explicit operator made the initialization of the atom's sparsity pattern roughly one hundred times faster.

The identity applies the operator; the atom must also store its Jacobian. The parent node receives $(I_p \otimes A)\, J$, where J is the Jacobian of the child, and transposes, indexes, and multiplies it like any other Jacobian. This matrix is made of p dense blocks at disjoint rows, one per column of the operand, and each block generally sits at its own global columns, so no single triple (X, r, c) describes it. Merging the blocks into one larger triple would store the full block diagonal, p copies of A separated by blocks of zeros, and falling back to generic sparse would lose the dense kernels. The engine instead keeps the blocks as they are, in a third representation.

Definition. A *stacked permuted dense* matrix is a vertical union of permuted dense blocks whose row vectors are pairwise disjoint; the column vectors may overlap. The blocks share one value buffer. Every kernel treats the blocks one at a time: structural atoms rewrite the index vectors of each block, and products distribute over the blocks, with one `dgemm` for each pair of blocks with intersecting row sets and nothing at all for the pairs with disjoint row sets, as in (3.2). The assembled matrix is never allocated or traversed, and each stacked kernel is a short loop over the corresponding permuted dense kernel. Figure 3.3 shows the Jacobian of AX for a variable X with two columns: one block per column of X, each a copy of A at its own rows and columns.

The three representations. Generic sparse, permuted dense, and stacked permuted dense are the only three matrix representations in the engine. Generic sparse is the default: it is produced by the variable leaves and by every atom that does not hold dense data. Permuted dense and stacked permuted dense are created by the two atoms that hold dense data, `left_matmul` and the quadratic form, and propagate from there through the structural atoms and the chain rule.

$$M = \begin{pmatrix} a_{00} & a_{01} & \cdot & \cdot & \cdot & \cdot \\ a_{10} & a_{11} & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & a_{00} & a_{01} & \cdot & \cdot \\ \cdot & \cdot & a_{10} & a_{11} & \cdot & \cdot \end{pmatrix}$$

$$X_1 = X_2 = A, \qquad r_1 = (0,1), \quad c_1 = (0,1), \qquad r_2 = (2,3), \quad c_2 = (2,3).$$

Figure 3.3 The Jacobian of AX as a stacked permuted dense matrix, for dense $A \in \mathbf{R}^{2\times 2}$ and a variable $X \in \mathbf{R}^{2\times 2}$ in a problem with six variables. There is one block per column of X, each a permuted dense copy of A at its own rows and columns. The 4×6 matrix M is never assembled; a product with M is one `dgemm` per block.

3.3 Computational kernels

A kernel is an operation on one or two of the matrix representations of §3.2: a transposition, a row selection, a chain-rule product. Every kernel is split into an alloc function and a fill function, called by the symbolic and numeric passes respectively: alloc computes the structure of the result and reserves its memory, and fill writes numerical values into the fixed structure. This section describes the kernels from the point of view of the atoms that call them. §3.3.1 describes how a product selects a kernel when each operand can have any of the three representations, and §3.3.2 collects the work that is done once during the symbolic pass instead of at every evaluation.

3.3.1 Kernel dispatch

Unary operations are dispatched through the function pointer table of §3.2, based on the representation of their single operand. The chain-rule products $B^T A$ and $B^T D A$ depend on the representations of both operands, which a single table cannot express. They are instead resolved with double dispatch: a dispatch function switches on the type tags of B and then of A, and calls one of nine kernels. Figure 3.4 shows the representation of the result for the nine combinations of operand types.

A product with a permuted dense or stacked permuted dense operand produces a result of the same kind, so once a dense block has been identified, it remains in a structured representation through all later products. Eight of the nine kernels are based on dense BLAS calls; in the mixed combinations, the columns of the sparse operand are gathered into the dense computation. Only the product of two generic sparse matrices uses Gustavson's algorithm [55], with the symbolic and numeric phases discussed in §1.5.2.

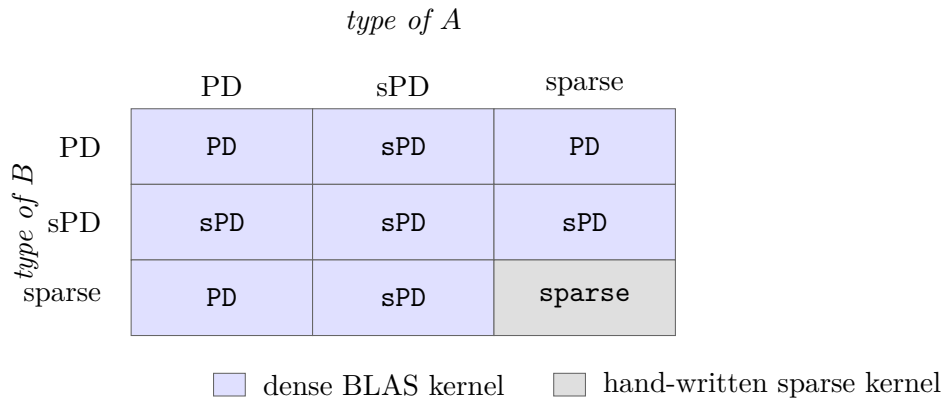


Figure 3.4 Double dispatch of the chain-rule products $B^T A$ and $B^T DA$. Each cell names the representation of the result for one combination of operand types (PD is permuted dense, sPD is stacked permuted dense); its color indicates the kernel family that computes it. Only the sparse-sparse combination falls back to a hand-written sparse kernel.

3.3.2 Reusing work across passes

Affine atoms. The Jacobian of an affine atom does not depend on x: for the structural atoms, such as indexing, transposition, reshaping, stacking, and summation, it is determined by shapes and index sets alone, and the remaining affine atoms, such as `left_matmul`, `scalar_mult`, `convolve`, and `kron`, place problem data in the Jacobian, whose values only change when a parameter update sets their refresh flag. The quadratic form is the only atom outside the affine class that appears in canonicalized problems; it contributes the objective matrix P through the Hessian oracle (§3.4).

Work done once. The numeric pass never allocates memory; every buffer, including the scratch space of the kernels, is allocated during the symbolic pass. Beyond allocation, several other kinds of work are done only once. Sparsity patterns are computed symbolically: `add` merges the patterns of its children with a sorted merge, the sparse products run the symbolic phase of Gustavson's algorithm [57], and the factored products of §3.2 determine their output index vectors before any values exist. Index maps are also precomputed: transposition stores its row permutation, `kron` its per-row copy maps, and `sum` the map from child nonzeros to output entries, so the numeric pass performs no index arithmetic. When the index vectors of two operands do not intersect, the product is recorded as empty during the symbolic pass, and its numeric pass does nothing. Finally, values that cannot change are written once; the Jacobian of a variable, for example, is filled during the symbolic pass and never touched again.

Figure 3.5 illustrates this for the `sum` atom. During the symbolic pass, the representation of the child allocates the output and fills `idx_map`, an array that maps each nonzero of the child Jacobian to the output entry it is added to. The axis convention and all other index decisions are handled at this point. The numeric pass is a single loop, the same for every axis and every representation. Indexing, transposition, and `kron` work the same way, with a copy instead of an addition.

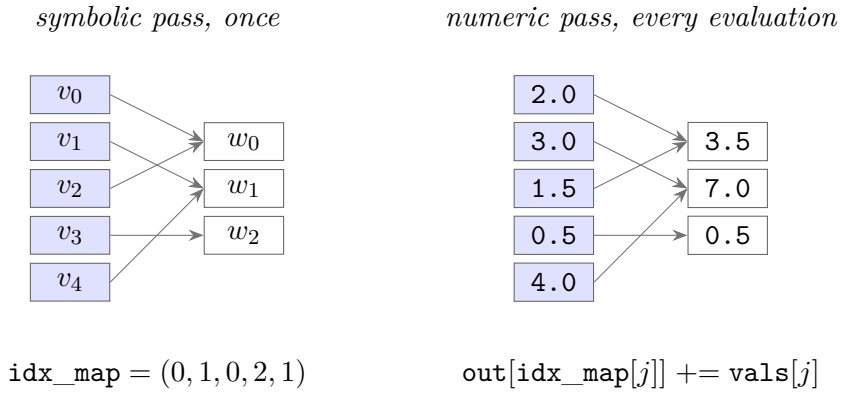


Figure 3.5 The two passes of the `sum` atom. During the symbolic pass, `idx_map` is built, shown as arrows from the nonzeros v_j of the child Jacobian to the output entries w_i; at every evaluation, the values are accumulated along the fixed arrows. The numeric passes of the other structural atoms have the same form.

The problem level. Assembly works in the same way. The solver receives one global CSR matrix, formed by concatenating the constraint Jacobians, whose layout is fixed during the symbolic pass; an evaluation only copies the values of each constraint root into this matrix at a fixed offset. Affine constraints are skipped entirely after the first evaluation, since their values cannot change until the parameters do.

Parameter updates. An update only involves the numeric pass: `update_params` writes the new values into the parameter leaves and sets a refresh flag on each affected subtree, and the next evaluation recomputes the values through the fixed maps and kernels. In comparison, without parametric compilation [25], `CVXPY` traverses the expression graph again, recomputes every sparsity pattern, and reallocates every buffer each time a parameter changes. The difference is measured in Chapter 4.

3.4 The API and the CVXPY interface

The engine is distributed as a C library with a thin Python binding called `sparsediffpy`, which exposes the node constructors. It is used in two places inside `CVXPY`. The nonlinear programming interface wraps the bindings in a class called `C_problem`, which is built from the expression trees of a problem. The canonicalization backend, which is the one measured in this thesis, is selected by passing `ignore_dpp=True`, and replaces the stuffed matrices of §2.1 with a cone program object that wraps the same `C_problem`. The code below is lightly condensed from the actual interface.

The lifecycle of a `C_problem` mirrors the symbolic and numeric passes:

```
prob = C_problem(problem)     # convert the tree, register parameters
prob.init_jacobian_coo()      # symbolic pass: patterns, maps, buffers
rows, cols = prob.get_jacobian_sparsity_coo() # structure, fetched once


prob.objective_forward(x)     # numeric pass, repeated per iterate
prob.constraint_forward(x)
grad = prob.gradient()
vals = prob.eval_jacobian_vals() # values aligned with (rows, cols)
```

The sparsity pattern is fetched once and does not change afterwards; the later calls return value arrays that are aligned with it.

The canonicalization backend extracts the problem data, an objective $\frac{1}{2}x^TPx + q^Tx + d$ and constraints $Ax + b$ that must lie in a cone, by evaluating the oracles at $x = 0$. The forward passes give d and b, the gradient gives q, and the Jacobian and Hessian values fill the fixed patterns of A and P. Parameter updates go through a single method:

```
def apply_parameters(self, id_to_value):
    theta = flatten_and_concatenate(id_to_value)
    if theta == self.cached_theta:    # same values: reuse everything
        return self.P, self.q, self.d, self.A, self.b
    self.c_problem.update_params(theta) # numeric pass only
    self.P, self.q, self.d, self.A, self.b = self.extract() # at x = 0
    self.cached_theta = theta
    return self.P, self.q, self.d, self.A, self.b
```

Repeated solves with unchanged parameter values return the cached matrices; an actual update costs one run of the numeric pass.

The caching follows the same pattern as in §3.3.2. The `C_problem` and its sparsity structures

live as long as the problem family, the extracted matrices are kept until the parameter values change, and only numerical values are recomputed. After construction, nothing in the Python layer traverses an expression tree. Chapter 4 compares the engine with `CVXPY`'s canonicalization backends, implementations of the matrix-stuffing approach reviewed in Chapter 2.

References. The row-wise sparse product and the symbolic and numeric split are due to Gustavson [55]; see also [57]. Mixing dense blocks into sparse storage appears in several settings: supernodal and multifrontal solvers gather dense submatrices of a sparse factor and process them with level-3 BLAS [83, 84] (see also the survey [85]), and blocked sparse formats and autotuned kernels exploit small dense blocks found in the matrix [86, 87]. In these systems the blocks are discovered in the data; in the engine they are known from canonicalization.

CHAPTER 4 BENCHMARK RESULTS

This chapter presents benchmark results which compare the `diffengine` backend (the `CVXPY` implementation based on `SparseDiffEngine`) against `CVXPY`'s own canonicalization backends: the default `cvxcore` backend, the matrix-stuffing baseline of §2.1, and the two pure-Python backends, `SCIPY` and `COO`, that `CVXPY` ships as alternatives to it. The comparison is restricted to `CVXPY`'s own pipeline because that is the claim under test: `diffengine` is a drop-in replacement for the canonicalization backend, and the relevant question is whether it constructs the same problem data faster. All backends run on the 25 problems of `CVXPY`'s public benchmark suite (`https://github.com/cvxpy/benchmarks`). These convex optimization problems cover many fields, ranging from portfolio models, robust regression, and semidefinite programs to simple LPs and QPs that are close to standard form. The rest of the chapter first presents some technical details about the benchmark setup, followed by the comparison results which are paired with commentary and observations.

4.1 Benchmark Setup

Every benchmark measures the construction of the problem data (A, b, c, P); no solver is invoked. For both backends this corresponds to a call to `get_problem_data` with the target solver fixed to `Clarabel` [23]: the `CPP` (`cvxcore`) baseline runs on the default DPP path, while `diffengine` is selected by passing `ignore_dpp=True`. The alternative `SCIPY` and `COO` backends of §4.3 are selected through the `canon_backend` argument of the same call.

Every comparison reports up to two timings: a *single* solve, where the problem data are constructed once from scratch, and, on the subset of parametric problems, a *re-solve*, where the problem data are reconstructed after reassigning random parameter values on a cached problem.

All measurements run single-threaded (BLAS thread pools pinned to one thread) on a 2018 MacBook Pro with a 6-core Intel Core i7-8850H at 2.6 GHz and 16 GB of RAM. The software versions are:

- `Python` 3.14,
- `CVXPY` 1.10 (development version),
- `sparsediffpy` 0.6.1,

- `NumPy` 2.4 [39] and `SciPy` 1.17 [58].

Each cold compilation is measured once, in a fresh process per backend so that no cached compilation chain is reused; the re-solve columns are means of three re-evaluations. The `cvxcore` and `diffengine` timings were recorded in one session; the `SCIPY` and `COO` timings of §4.3 were recorded in a second session on the same machine and software versions, in which re-measured `cvxcore` times reproduced the first session within a few percent. Before any timing, `diffengine` is verified to reproduce the `cvxcore` problem data bit-identically ($\max|\Delta| = 0$) at full benchmark size. The scripts which run the benchmarks and all recorded timings are available at `https://github.com/SparseDifferentiation/Jacobian-accumulation` under the `canonicalization/` directory.

4.2 Single-solve compilation against `cvxcore`

A fresh problem is formed and canonicalized once from scratch (Table 4.1).

`diffengine` is faster on 24 of the 25 problems, with speedups between 1.2× and 124× and a geometric mean of 4.2×. The largest speedups occur where `cvxcore`'s per-expression tensor construction dominates the compilation time, as in the portfolio and regression models with many scalar constraints. UnconstrainedQP, a tiny problem wrapped in a large `kron` expression, is the extreme case at 124×. The speedups shrink on problems whose cost is a single large data copy, such as OptimalAdvertising and CVaR. Finally, the compilation of ConvexPlasticity, is dominated by `CVXPY`'s Python-side reductions rather than by either backend.

4.3 Single-solve compilation against the `SCIPY` and `COO` backends

Beyond the default C++ `cvxcore`, `CVXPY` ships two pure-Python canonicalization backends built on `scipy.sparse` [58]. The `SCIPY` backend represents each slice of the parameter tensor as a two-dimensional sparse matrix, while the more recent `COO` backend (introduced in 2025) keeps the whole tensor in coordinate format (§1.5.2) so that its operations scale with the number of stored nonzeros even when the parameters are large. Re-running the cold-compile experiment of §4.2 under these two backends answers whether the conclusions above depend on the choice of `cvxcore` as the baseline (Table 4.2). The `CPP` and `diffengine` columns repeat those of Table 4.1.

The sweep first shows that `cvxcore` is not uniformly the fastest of `CVXPY`'s own backends. The pure-Python `SCIPY` backend beats `CPP` on 18 of the 22 completed problems, usually by

Table 4.1 Cold compilation time across all 25 benchmarks, `diffengine` versus `cvxcore`. The ratio is `diffengine`/`CPP`; values below 1 favor `diffengine`. The final row is the geometric mean of the per-problem ratios.

Benchmark	`CPP` (s)	`diffengine` (s)	Ratio
UnconstrainedQP	9.32	0.08	**0.01**
Cajas	19.66	1.06	0.05
FactorCovarianceModel	5.24	0.73	0.14
SVMWithL1Regularization	11.16	1.60	0.14
Yitzhaki	3.07	0.47	0.15
HuberRegression	10.97	1.61	0.15
LeastSquares	8.87	1.42	0.16
SimpleQP	10.49	1.69	0.16
SimpleScalarParametrizedLP	5.29	0.92	0.17
SimpleLP	30.07	5.29	0.18
SDPSegfault1132	59.75	12.47	0.21
SimpleFullyParametrizedLP	2.12	0.45	0.21
ParametrizedQP	10.19	2.25	0.22
TvInpainting	3.57	1.06	0.30
SlowPruning	6.88	2.15	0.31
SemidefiniteProgramming	5.38	1.93	0.36
ConeMatrixStuffing	10.34	3.76	0.36
QuantumHilbertMatrix	3.89	1.64	0.42
SmallMatrixStuffing	6.52	2.77	0.42
ParamSmallMatrixStuffing	0.48	0.21	0.43
ParamConeMatrixStuffing	0.31	0.14	0.45
Murray	5.79	3.43	0.59
CVaR	34.80	25.31	0.73
OptimalAdvertising	6.06	5.20	0.86
ConvexPlasticity	2.09	2.13	**1.02**
Geometric mean			**0.24**

Table 4.2 Cold compilation time under all three `CVXPY` canonicalization backends, compared with `diffengine`. The ratio is `diffengine` over the best of the three backends on each problem; values below 1 mean `diffengine` beats all of them. The final row is the geometric mean of the per-problem ratios over the 22 problems on which every backend completed.

Benchmark	`CPP` (s)	`SCIPY` (s)	`COO` (s)	`diffengine` (s)	Ratio
UnconstrainedQP	9.32	6.47	12.12	0.08	**0.01**
HuberRegression	10.97	10.07	10.61	1.61	0.16
LeastSquares	8.87	9.16	10.18	1.42	0.16
SimpleQP	10.49	10.17	11.23	1.69	0.17
Yitzhaki	3.07	2.80	3.11	0.47	0.17
FactorCovarianceModel	5.24	4.21	5.04	0.73	0.17
SVMWithL1Regularization	11.16	8.78	9.53	1.60	0.18
SimpleLP	30.07	28.92	27.23	5.29	0.19
SimpleScalarParametrizedLP	5.29	4.33	4.52	0.92	0.21
SlowPruning	6.88	6.85	7.58	2.15	0.31
TvInpainting	3.57	3.15	3.31	1.06	0.34
Cajas	19.66	2.92	3.17	1.06	0.36
ConeMatrixStuffing	10.34	9.44	16.49	3.76	0.40
SmallMatrixStuffing	6.52	7.20	10.28	2.77	0.42
QuantumHilbertMatrix	3.89	3.80	4.47	1.64	0.43
ParamSmallMatrixStuffing	0.48	27.96	0.42	0.21	0.49
ParamConeMatrixStuffing	0.31	7.45	0.27	0.14	0.52
SemidefiniteProgramming	5.38	2.31	3.53	1.93	0.84
Murray	5.79	3.73	3.87	3.43	0.92
CVaR	34.80	27.32	26.68	25.31	0.95
SDPSegfault1132	59.75	10.12	40.14	12.47	1.23
OptimalAdvertising	6.06	2.18	2.08	5.20	**2.50**
ConvexPlasticity	2.09	`ValueError`		2.13	—
SimpleFullyParametrizedLP	2.12	crash		0.45	—
ParametrizedQP	10.19	>1200 s		2.25	—
Geometric mean					**0.33**

a modest 1.0–1.3× but by 5.9× on SDPSegfault1132 and 6.7× on Cajas. Its weakness is the parameter-heavy problems, where building one tensor slice per parameter entry collapses: ParamSmallMatrixStuffing takes 28.0 s against `CPP`'s 0.48 s. The `COO` backend repairs exactly those rows (0.42 s and 0.27 s, on par with `CPP`), which is what its coordinate-format tensor is designed for, at the price of trailing `SCIPY` elsewhere (SDPSegfault1132, ConeMatrixStuffing). Neither Python backend completes the full suite: both raise a `ValueError` on ConvexPlasticity, the sweep lost SimpleFullyParametrizedLP to a crash of the benchmark process, and ParametrizedQP exceeded the 1200 s per-problem timeout.

Measured against an oracle that picks the fastest of the three backends on each problem, `diffengine` remains the fastest option on 18 of the 22 completed problems and is within 10 % of the best on two more (Murray, CVaR), with a geometric-mean ratio of 0.33, that is, 3.1× faster than the oracle. It is clearly beaten only on SDPSegfault1132 (1.2× slower than `SCIPY`) and OptimalAdvertising (2.5× slower than `COO`); as in §4.2, both are problems whose compilation is dominated by moving one large block of constant data into the problem matrices rather than by computing coefficients. The conclusions of §4.2 therefore do not hinge on `cvxcore` being the baseline.

4.4 Parametric compilation against `cvxcore`

On the subset of parametric problems, we report both the first solve and the subsequent re-solve time after reassigning random parameter values (Table 4.3).

`diffengine` is faster on the re-solve for six of the eight parametric problems, and on the first solve for all but ConvexPlasticity. Its re-solves are fast because the compiled program is cached, so only the parameter-dependent coefficient entries are re-evaluated. The DPP baseline instead multiplies its cached parameter tensor by the new parameter vector, which is hard to beat when that tensor is small and dense. This explains the two rows where DPP remains around 2× faster: a scalar parameter whose effect is spread across many coefficients (SVM's regularization weight, FactorCovariance's risk aversion) is the ideal case for the tensor product.

Table 4.3 Parametric compilation, `diffengine` versus `cvxcore`: first solve and warm re-solve. The ratio is `diffengine`/`CPP` on the re-solve; values below 1 favor `diffengine`. The final row is the geometric mean of the per-problem ratios.

		First solve (s)		Re-solve (s)		
Benchmark	#par	`CPP`	`diffengine`	`CPP`	`diffengine`	Ratio
ParamConeMatrixStuffing	3	0.31	0.14	0.0030	0.0010	**0.33**
ParamSmallMatrixStuffing	3	0.48	0.21	0.0048	0.0017	0.35
SimpleFullyParametrizedLP	1	2.12	0.45	0.230	0.123	0.54
ParametrizedQP	2	10.19	2.25	1.361	0.959	0.70
SimpleScalarParametrizedLP	1	5.29	0.92	0.330	0.251	0.76
ConvexPlasticity	3	2.09	2.13	0.668	0.666	1.00
FactorCovarianceModel	2	5.24	0.73	0.144	0.277	1.92
SVMWithL1Regularization	1	11.16	1.60	0.263	0.530	**2.01**
Geometric mean						**0.78**

CHAPTER 5 CONCLUSION

5.1 Summary of Works

5.2 Limitations

5.3 Future Research

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